Teen Distress Pilot

Two recent accounts offer competing explanations for the post-2010 rise in adolescent anxiety and depression: Jonathan Haidt's *The Anxious Generation* attributes the shift to smartphones and social media, while Peter Gray points to intensified schooling and declining autonomy. This pilot report asks how far large-scale public microdata can go toward separating those hypotheses before any restricted or licensed datasets are requested. Every figure and table is computed from public microdata (ACS, ATUS, MICS, PISA, GSHS, IPUMS International) and published survey series.

The four hypotheses

The study is organized around four candidate explanations, with each empirical test designed to distinguish between at least two of them.

Hypothesis	Claim	What it predicts in these data
H1 · Phones and social media (Haidt, Twenge)	Smartphone-based social media, arriving between 2010 and 2013, caused the rise in adolescent distress, particularly among girls.	Distress rises earlier where smartphones and fast mobile data arrive first; increases are much larger for girls than for boys; distress rises in developing countries as mobile internet spreads; heavy users report worse outcomes; school phone bans reduce distress.
H2 · School pressure (Gray)	Accountability testing, heavier homework loads, and reduced free play made schooling more stressful from around 2010 onward.	No difference by local mobile-network timing; girls affected somewhat more than boys; increases concentrated in the high-stakes high-school years (15–17) and during the school year; distress rises where perceived schoolwork pressure rises; state accountability reforms (Race to the Top, Common Core testing) predict state outcomes.
H3 · Interaction	Smartphones are most harmful when combined with high-stakes schooling—social comparison and notifications running through the same school hours that carry tests and grades—so that removing phones from the school day brings disproportionate relief.	Both academic pressure and phone use rise after 2010; mobile-rollout effects are larger in high-pressure settings; school phone bans improve well-being, especially for academically anxious students; out-of-school screen use matters less.

Hypothesis	Claim	What it predicts in these data
H0 · Reporting or measurement drift	The observed rise largely reflects changing diagnostic thresholds, mental-health awareness, or survey reporting norms.	Older adults (including the 65+ comparison group) trend upward in the same way; behavioral, clinical, and self-report series diverge; no geographic or dose-response pattern appears.

Summary: what the public microdata support today

Each row summarizes one empirical test from the pilot, grouped by the hypothesis it addresses most directly. Full specifications, data sources, and caveats follow in Parts 1–4.

What was measured	What the pilot found	Relevant to
Is there a real rise, and is the outcome usable? (H0)		
65+ comparison group and adult trends	The ACS cognitive-difficulty rate falls steadily for adults 65+ (12.2% → 9.4% between 2008 and 2019) and is flat at ages 50–64 while rising for all age groups under 50, ruling out generic survey-wide response drift.	against H0
Agreement across measures	Clinical depression interviews (NSDUH), persistent sadness (YRBS), and the ACS cognitive-difficulty item all turn upward between 2011 and 2013, with girls rising faster in every source; within-state changes move together. However, the ACS item captures only about a sixth of the proportional rise in depression.	real rise; ACS usable but weak
Subgroup gradient	Relative increases in the ACS item are steepest among affluent, private-school, Hispanic, Asian, and foreign-born teens, and smallest among low-income teens—a pattern consistent with both early smartphone adoption and differential diagnosis-seeking.	ambiguous
Phones and social media (H1)		
Sex pattern	Girls' relative increase exceeds boys' across all sources: ACS +17% vs. +9% (2008–10 → 2017–19), YRBS sadness +26% vs. +19%, and post-2020 ACS increases of +45–55% for girls vs. +16–23% for boys.	H1 / H3
Timing in early vs. late LTE markets	No rollout-timing effect. Early- and late-LTE metropolitan areas track each other closely through 2013; the static DiD estimate is +0.02 pp (SE 0.15) across 279 dated metros and −0.04 pp (SE 0.19) for girls, with similarly null results across county panels in six state youth surveys.	not H1 via rollout timing
U.S. smartphone adoption	State-level teen-household smartphone-internet share predicts YRBS girls' sadness within states (+0.90 pp per 10 points, SE 0.44), but metro adoption does not predict sadness at the county level (+0.13 pp overall across five states, SE 0.13; +0.22 pp for girls in California and Maine, SE 0.25).	weak H1

What was measured	What the pilot found	Relevant to
Phone-first countries, adult- or cluster-measured exposure	MICS (67 surveys with valid responses): caregiver-reported anxiety and depression show no meaningful relation to local young-adult connectivity or a teenage girl's own daily phone/internet use; the small positive cluster correlation for girls' anxiety also appears among 5–9-year-olds. GSHS (34 countries): no within-country relation to national internet users, mobile-broadband subscriptions, or 3G/4G coverage. Census cognitive-difficulty items in Kenya and Mexico are uninformative.	not H1 on adult- or cluster-measured use
Adolescent self-reported phone hours	GSHS 2022–23 (five countries): girls using a phone 3+ hours/day report loneliness +8.2 pp and suicidal ideation +10.1 pp higher than lighter users, rising monotonically with hours; effects for boys are roughly half as large. Nepal 2022 DHS PHQ-9/GAD-7 self-reports point in the same direction for 15–19-year-old women. Cross-sectional self-reports, leaving causal direction open.	H1 pattern, not timing
School pressure (H2)		
Age pattern and time use	Through 2019, the ACS increases at ages 12–14 and 15–17 are similar and 18–24 rises faster, showing no high-stakes high-school signature (seasonality is not available in public ACS files). In ATUS, school-weekday homework time rose nearly 40% (from ~50 to ~71 minutes) between 2003–08 and 2015–19 while sleep held steady and in-person socializing fell.	school demand rose (H2 premise)
HBSC schoolwork pressure	Perceived schoolwork pressure and psychosomatic complaints both rose after 2009/10 across ~45 countries, and complaints rose more where pressure rose more (slope 0.14–0.24). However, the within-country slope is identical in the 2002–2010 baseline period, and most of the post-2010 increase is common across countries.	H2 partly pressure is a stable correlate, not the break
State accountability policy	ESEA waivers, Common Core–aligned testing, test-based teacher evaluation, and exit exams show no positive relation to ACS, YRBS, or NSDUH distress; the apparent static Race to the Top coefficient disappears in an event study around the award year.	not H2 via U.S. policy shocks
Interaction (H3)		
LTE × school pressure	Early-LTE metros inside high-accountability states did not experience a larger rise in the ACS item; where identified, the interaction point estimates for girls are negative.	not H3 via state policy
School phone bans	PISA 2022 (80 systems): school phone bans reduce daily in-school smartphone use by 12.1 pp and reported classroom distraction by 3.5 pp; girls in ban schools report 1.4 pp less loneliness, 2.0 pp less device anxiety, and +0.02–0.03 SD higher belonging and life satisfaction. Cross-sectional comparison across schools. Nationally, U.S. school prohibitions fell from 91% to 66% during 2009–16 and re-tightened after 2016 as sadness rose and then fell.	H1 / H3 direction, small; U.S. test pending

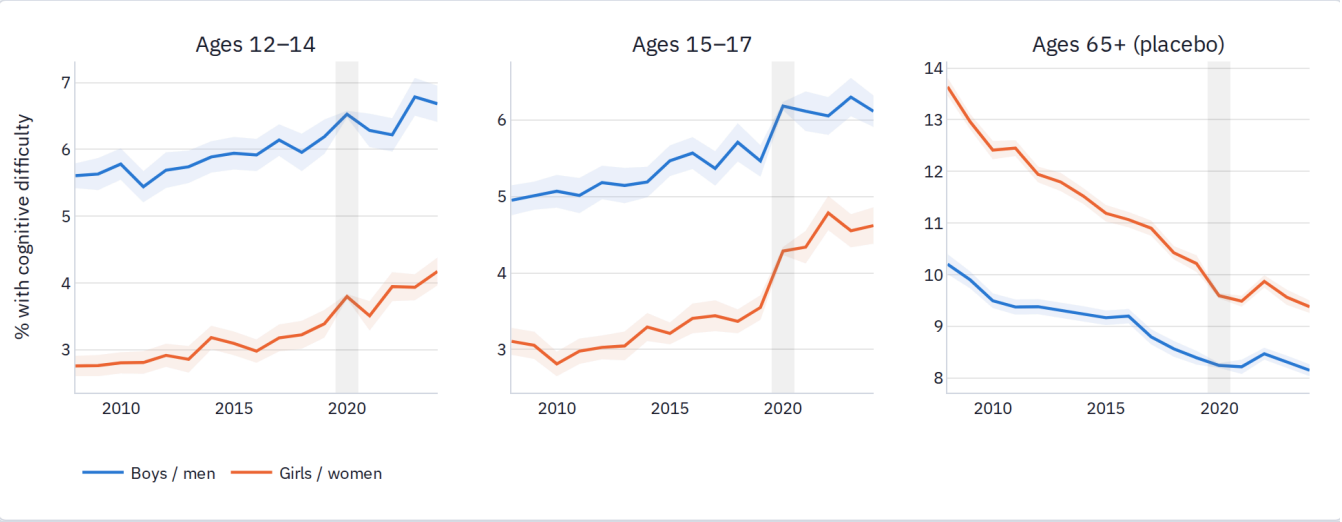
What was measured	What the pilot found	Relevant to
Bans × academic pressure	PISA 2022: the association between school phone bans and lower loneliness is larger for students with above-median mathematics anxiety —the available proxy for academic pressure—(girls −1.15 vs. −0.27 pp; boys −0.81 vs. +1.12 pp), though the interaction is imprecisely estimated.	H3 direction, not established

Part 1 · The rise, and whether public data can measure it

Testing either hypothesis requires first establishing the outcome in public data: verifying that adolescent distress rose after 2010, checking whether the increase reflects a genuine shift rather than survey reporting drift, and evaluating how closely the Census disability screener tracks clinical depression and self-reported sadness. This section combines the American Community Survey with national mental-health surveys (NSDUH, YRBS, NHANES) and county-level state youth surveys.

The ACS cognitive-difficulty item, 2008–2024

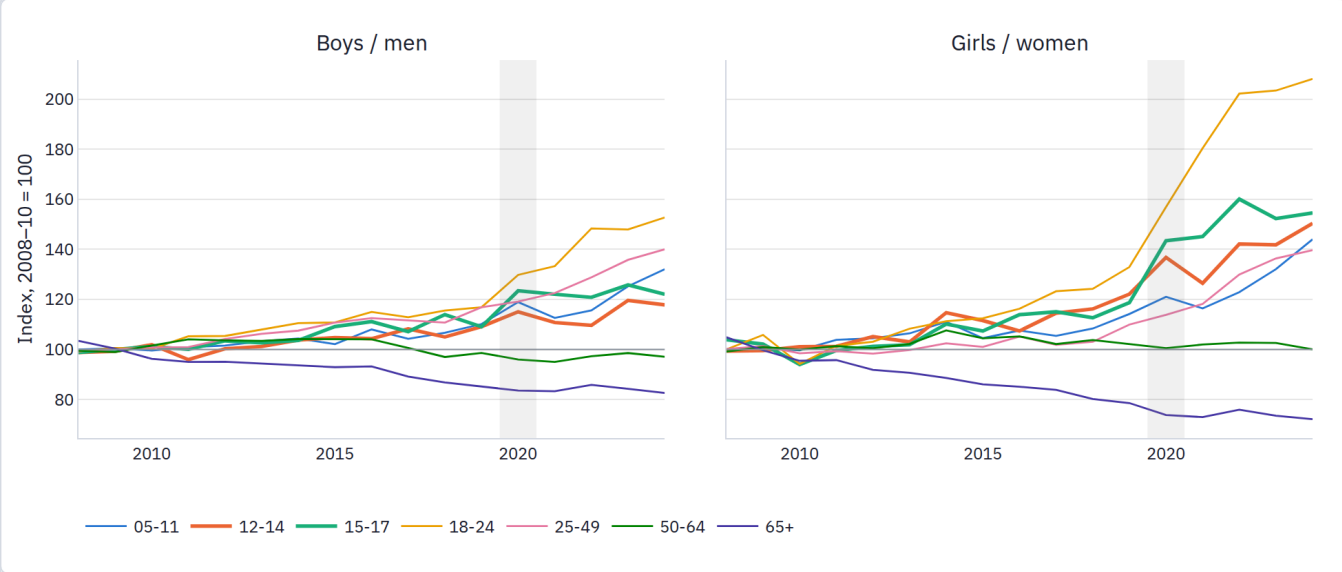
The American Community Survey asks of every household member aged 5 and older whether the person has "serious difficulty concentrating, remembering, or making decisions" because of a physical, mental, or emotional condition. The question wording has been unchanged since 2008 (2005–07 used different wording and is excluded). Each annual wave samples roughly 240,000 adolescents aged 12–17. All standard errors are computed using the 80 Census successive-difference replicate weights.



Shaded bands are 95% confidence intervals. The grey band marks the 2020 experimental ACS release, which suffered from pandemic-related nonresponse bias.

Age	Sex	2008–10 %	2017–19 %	Rel. change	2022–24 %	Rel. change
12-14	boys	5.67	6.10	+7%	6.56	+16%
12-14	girls	2.78	3.27	+18%	4.02	+45%
15-17	boys	5.01	5.51	+10%	6.16	+23%
15-17	girls	2.99	3.45	+15%	4.65	+56%

Age gradient



Each series plots the age group's rate indexed to its own 2008–10 mean (= 100). Thick lines highlight the adolescent age groups (12–14 and 15–17).

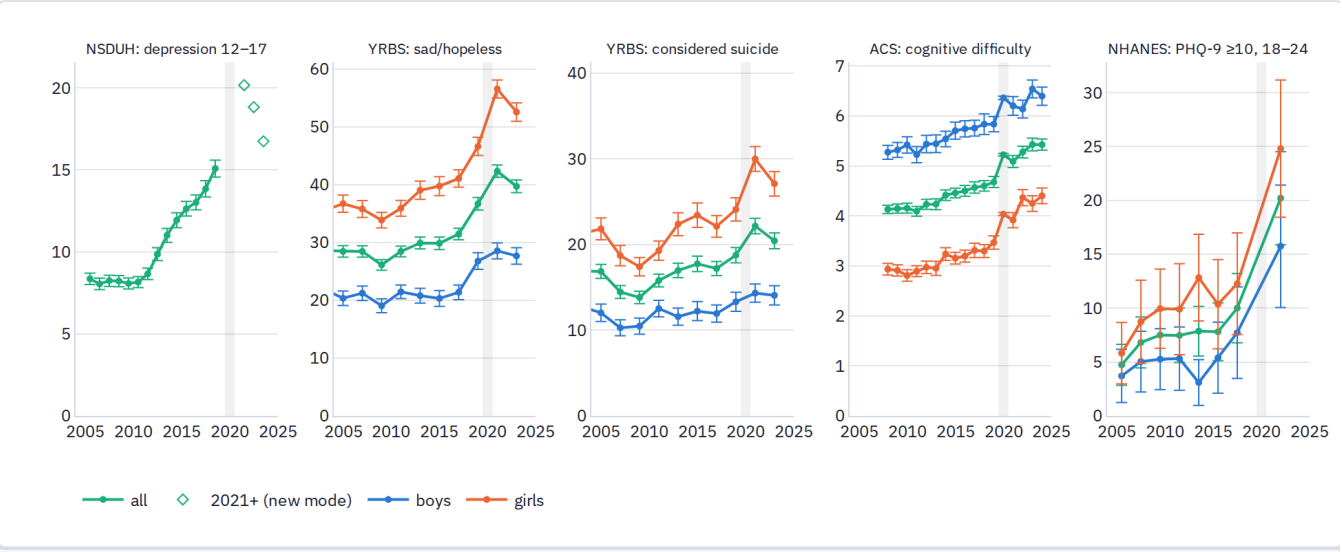
The ACS age series establish two basic facts. First, the pre-pandemic 2010s increase in cognitive difficulty is statistically clear but modest in absolute terms (about half a percentage point on a baseline of 3–5%), and it appears across all age groups under 50—rising fastest among 18–24-year-olds—while falling steadily among adults 65 and older. Second, the post-2020 surge is much steeper and is heavily concentrated among girls and young women. Because the ACS question is a severe disability screener, it will miss mood or anxiety disorders that do not impair daily concentration; the next section evaluates how well it tracks direct measures of adolescent depression.

Does the ACS item measure adolescent distress?

Before using the ACS cognitive-difficulty item as an outcome, we test it against three criteria: whether its national timing matches established adolescent depression and sadness surveys, whether its increase is concentrated among girls, and whether state-level changes in the ACS item move with state-level changes in clinical depression. Because the public NHANES depression

screener (PHQ-9) is released only for respondents aged 18 and older (the adolescent file is restricted), NHANES enters as an 18–24 benchmark.

National timing



All series in percent, with 95% CIs. NSDUH points sit at the midpoint of each two-year period; the 2021+ NSDUH estimates use a new web mode and DSM-5 criteria and are not directly comparable to earlier years. YRBS sad/hopeless was first asked in 1999.

NSDUH past-year major depressive episodes among 12–17-year-olds held steady at roughly 8% from 2005 through 2010–11 and then rose in nearly every two-year wave to 15% by 2018–19. YRBS persistent sadness was flat from 1999 through 2011 before rising through 2019. The ACS cognitive-difficulty series turns upward around 2012 and accelerates after 2019. It inflects at roughly the same time as the mental-health surveys, though its relative increase over the 2010s is much smaller—about +13% overall, compared with +85% in NSDUH.

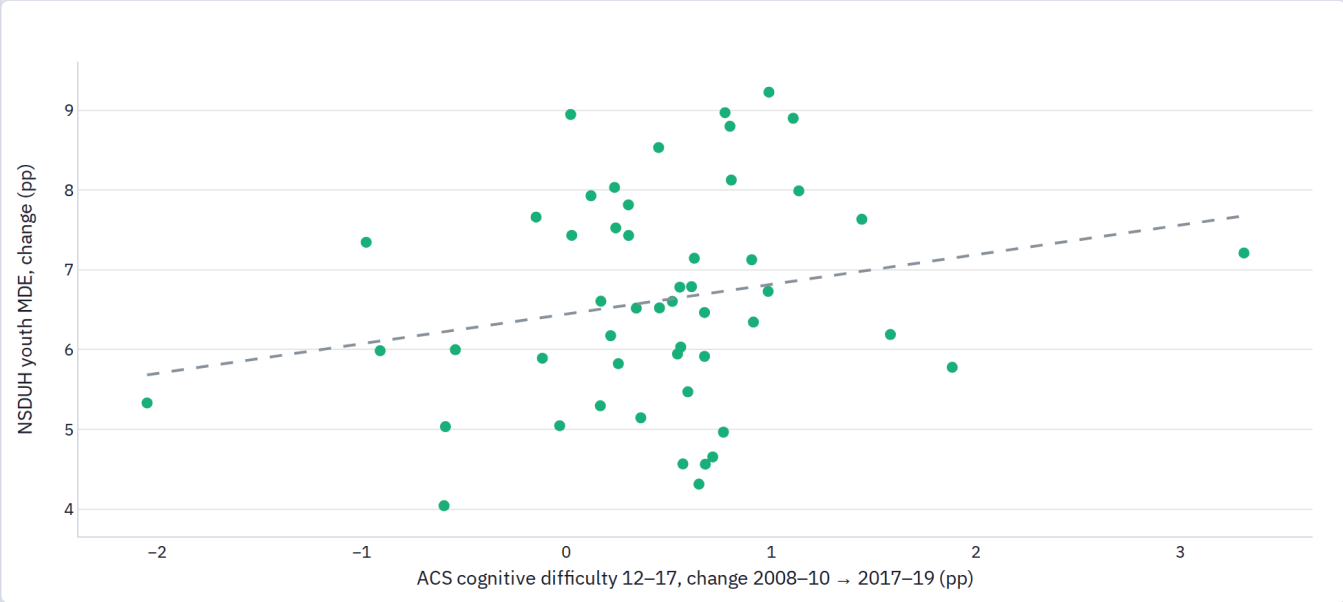
Sex pattern

Series	Sex	2009–11	2017–19	Relative change
ACS cognitive difficulty 12–17	boys	5.3	5.8	+9%
ACS cognitive difficulty 12–17	girls	2.9	3.4	+17%
YRBS sad/hopeless	boys	20.3	24.1	+19%
YRBS sad/hopeless	girls	34.9	43.9	+26%
YRBS considered suicide	boys	11.5	12.6	+10%
YRBS considered suicide	girls	18.3	23.1	+26%
NHANES PHQ-9 ≥10, 18–24	boys	5.3	7.7	+46%

Series	Sex	2009–11	2017–19	Relative change
NHANES PHQ-9 ≥10, 18–24	girls	9.9	12.3	+24%

Across all four sources, the relative increase is larger for girls than for boys. The ACS item passes this check: if the 2010s increase were driven primarily by childhood ADHD or learning-disability diagnoses, boys would dominate the change.

Do states move together?



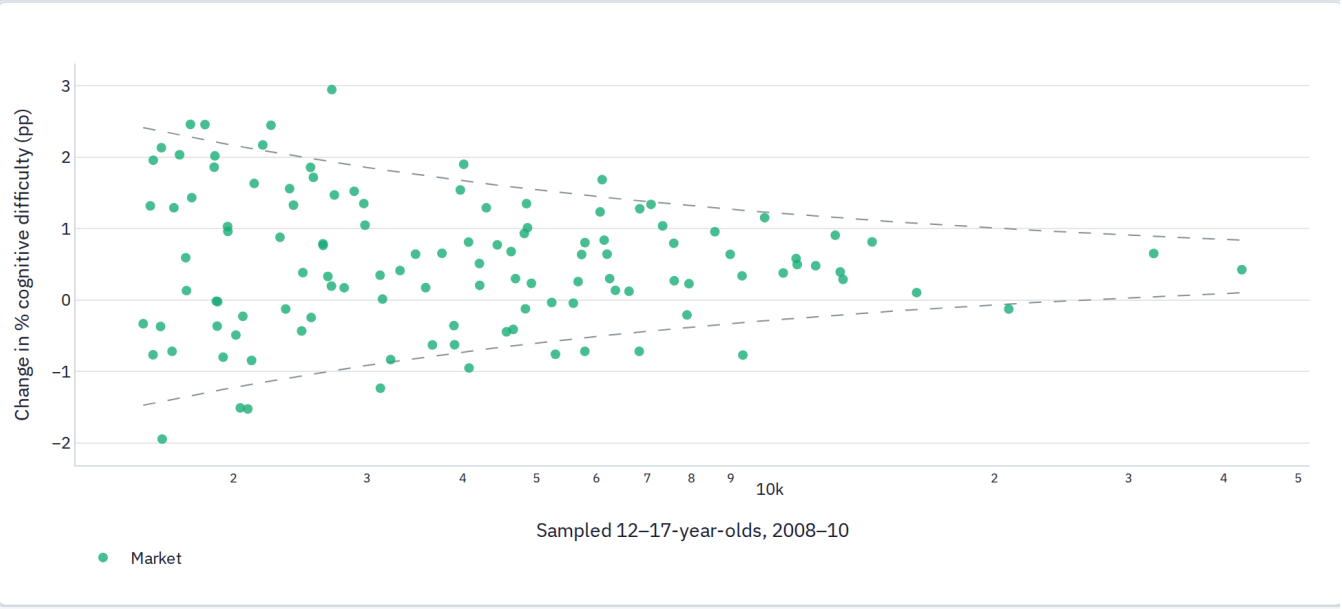
51 jurisdictions (50 states plus DC). Each point plots a state's change in both measures between 2008–10 and 2017–19; the dashed line is the OLS fit (slope 0.37, SE 0.18, r = 0.22).

In cross-section, state levels of the ACS item and NSDUH depression do not correlate, which is unsurprising given that one is a broad disability screener and the other is a diagnostic interview. Over time, however, their within-state changes move together: a state-and-period fixed-effects regression of NSDUH youth depression on the ACS cognitive-difficulty rate yields a slope of 0.14 (SE 0.05, p = 0.01) across 714 state-period observations, and the decade-long changes from 2008–10 to 2017–19 correlate at r = 0.22 (slope 0.37, SE 0.18). Against the YRBS state files (39 states, by sex), the ACS rate co-moves with persistent sadness at a slope of 0.35 (SE 0.18) with state-by-sex and year-by-sex fixed effects, though not with suicidal ideation.

Assessment. The ACS cognitive-difficulty item reproduces the female-led rise and co-moves with depression within states, making it usable for fine-grained geographic tests where no mental-health survey exists. Even so, it captures only about a sixth of the proportional rise in adolescent depression, and its girl-specific change is small relative to metro-level sampling error. For that reason, we pair every U.S. ACS test with state and county youth mental-health surveys.

Geographic variation: is there anything to explain?

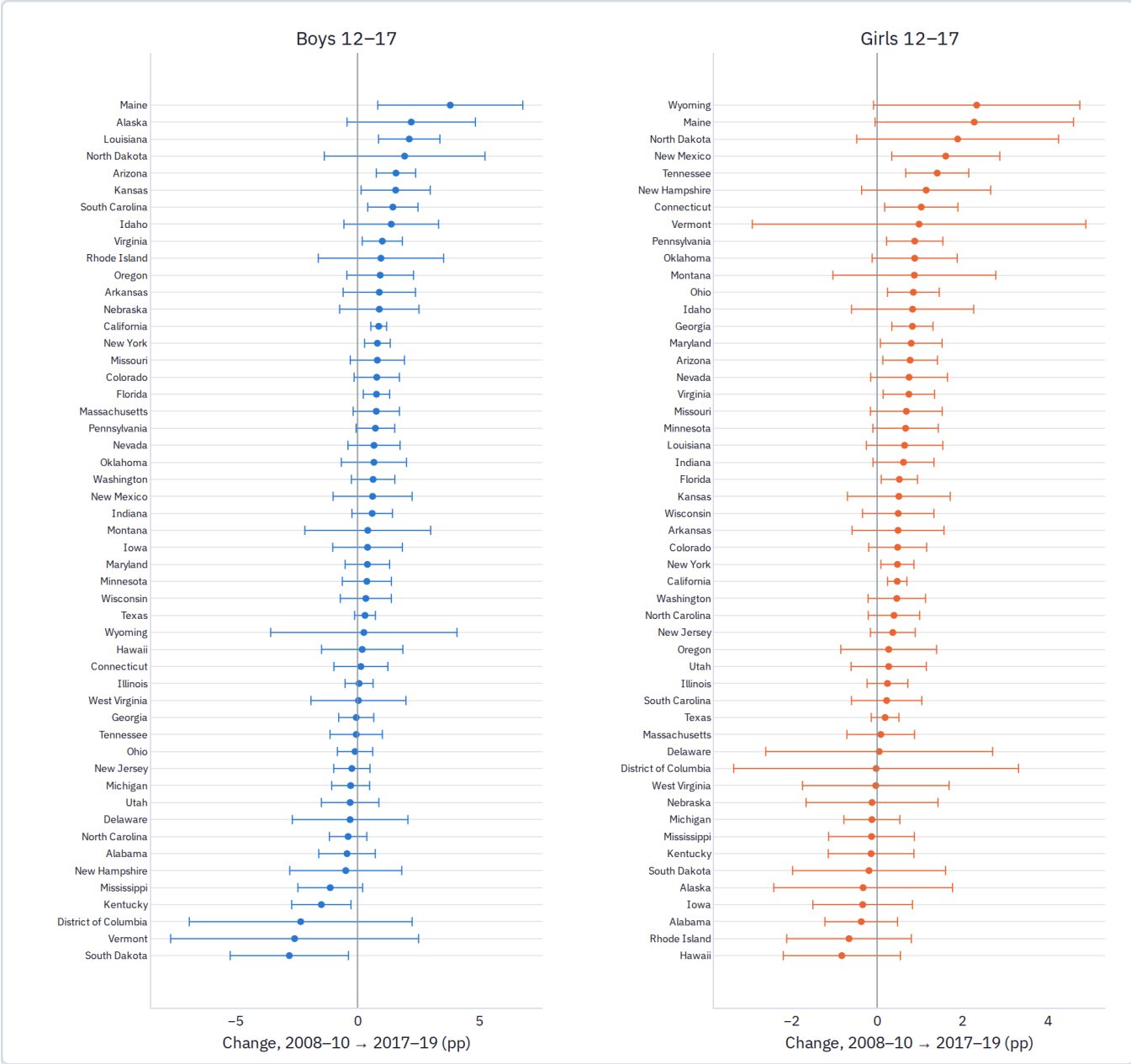
For a geographic rollout design to have statistical power, 2010s changes in adolescent distress must vary across local markets by more than sampling noise. Grouping the ACS into metropolitan areas (plus each state's non-metro remainder), the weighted mean change from 2008–10 to 2017–19 among 12–17-year-olds is +0.47 percentage points, the estimated true between-market standard deviation is 0.46 points, and the median market-level sampling error is 0.64 points (Cochran's Q = 218 on 116 df, $I^2 \approx 0.47$). Real cross-market differences exist, but individual metro estimates are noisy and must be pooled in regression models.



117 markets with at least 1,500 sampled 12–17-year-olds in 2008–10. Dashed curves show the ±1.96 SE funnel around the pooled mean for a market of a given sample size.

States, by sex

Disaggregating state-level changes by sex reveals a major constraint for U.S. geographic designs: while boys' changes vary substantially across states ($I^2 \approx 0.46$), girls' changes are nearly uniform nationwide ($I^2 \approx 0.05$). Because the female series is the component that reflects internalizing distress, there is very little cross-state variation in the ACS for either a phone-rollout or a school-policy design to exploit.



Who is driving the rise

Relative changes by demographic and socioeconomic subgroup, ages 12–17. The largest proportional increases occur among adolescents in higher-income families and private schools.

By family income relative to poverty line

Group	Sex	2008–10 %	2017–19 %	Rel. change to 2017–19	Rel. change to 2022–24	Share of 12–17, 2008–10
100-199%	boys	6.67	7.05	+6%	+15%	11%
100-199%	girls	3.50	4.30	+23%	+46%	10%

Group	Sex	2008–10 %	2017–19 %	Rel. change to 2017–19	Rel. change to 2022–24	Share of 12–17, 2008–10
200-399%	boys	4.55	5.17	+14%	+31%	16%
200-399%	girls	2.38	2.91	+22%	+78%	16%
400%+	boys	2.95	3.73	+26%	+48%	16%
400%+	girls	1.55	2.04	+32%	+89%	15%
poor <100%	boys	8.61	9.02	+5%	+11%	8%
poor <100%	girls	4.88	5.45	+12%	+36%	8%

By school type

Group	Sex	2008–10 %	2017–19 %	Rel. change to 2017–19	Rel. change to 2022–24	Share of 12–17, 2008–10
not in school	boys	8.32	8.54	+3%	+15%	1%
not in school	girls	6.21	5.82	-6%	+25%	1%
private	boys	4.53	5.91	+30%	+44%	5%
private	girls	2.57	3.61	+40%	+70%	5%
public	boys	5.33	5.71	+7%	+17%	45%
public	girls	2.82	3.26	+16%	+50%	43%

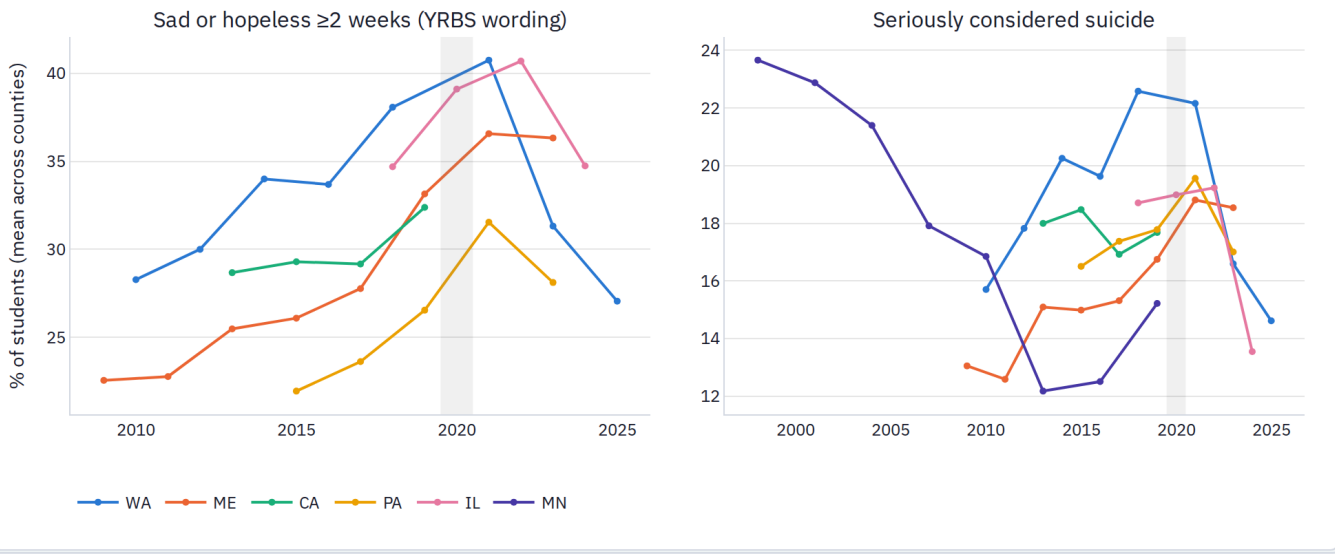
By race / ethnicity

Group	Sex	2008–10 %	2017–19 %	Rel. change to 2017–19	Rel. change to 2022–24	Share of 12–17, 2008–10
Asian-PI NH	boys	1.66	2.24	+35%	+90%	2%
Asian-PI NH	girls	1.10	1.47	+33%	+107%	2%
Black NH	boys	6.67	7.14	+7%	+13%	7%
Black NH	girls	3.69	3.94	+7%	+26%	7%
Hispanic	boys	4.09	5.16	+26%	+48%	10%
Hispanic	girls	2.37	2.95	+24%	+74%	10%

Group	Sex	2008–10 %	2017–19 %	Rel. change to 2017–19	Rel. change to 2022–24	Share of 12–17, 2008–10
Other-multi NH	boys	7.64	7.96	+4%	-8%	2%
Other-multi NH	girls	4.50	4.73	+5%	+7%	2%
White NH	boys	5.54	5.88	+6%	+12%	29%
White NH	girls	2.87	3.44	+20%	+51%	28%

State youth surveys at county level

To measure self-reported adolescent distress below the state level, we compiled county-level school survey tables from six states: Minnesota (1998–2019, 87 counties), Washington (2010–2025, 38 counties), Maine (2009–2023, 16 counties), California (2013–2019, 55 counties, by sex), Pennsylvania (2015–2023, 60 counties), and Illinois (2018–2024, 82 counties for sadness and 75 for suicidal ideation), totaling 34,439 county-grade-year cells with documented question wording and primary sources. Minnesota's pre-2013 sadness question asked about feeling sad over the past 30 days rather than the standard YRBS two-week item and is excluded from the sadness panel.



Unweighted county means across the grade levels each state reports in every survey wave (Washington grades 8, 10, and 12; Maine's 9–12 and Pennsylvania's 6–12 all-grade aggregates; Minnesota grade 9; California grades 7, 9, and 11 for sadness and 9 and 11 for suicidal ideation; Illinois grades 8, 10, and 12 for sadness and 10 and 12 for suicidal ideation). Minnesota's 2022 county tables are behind a web portal CAPTCHA and are not yet included.

The county panels highlight three notable patterns. First, persistent sadness rose across every state from roughly 2013 to a peak in 2021 (for example, from 28% to 41% in Washington, 23% to 37% in Maine, and 35% to 41% in Illinois). Second, Minnesota's 9th-grade suicidal-ideation series

—the only county series extending back to the 1990s—*fell* steadily from 24% in 1998 to 12% in 2013 before turning upward to 15% by 2019. Third, the post-2021 waves show a substantial downturn: Washington's 2025 sadness rate (27%) is below its 2010 baseline, and Illinois's 2024 suicidal-ideation rate (13.5%) is the lowest in its county series.

Part 2 · Phones and social media (H1)

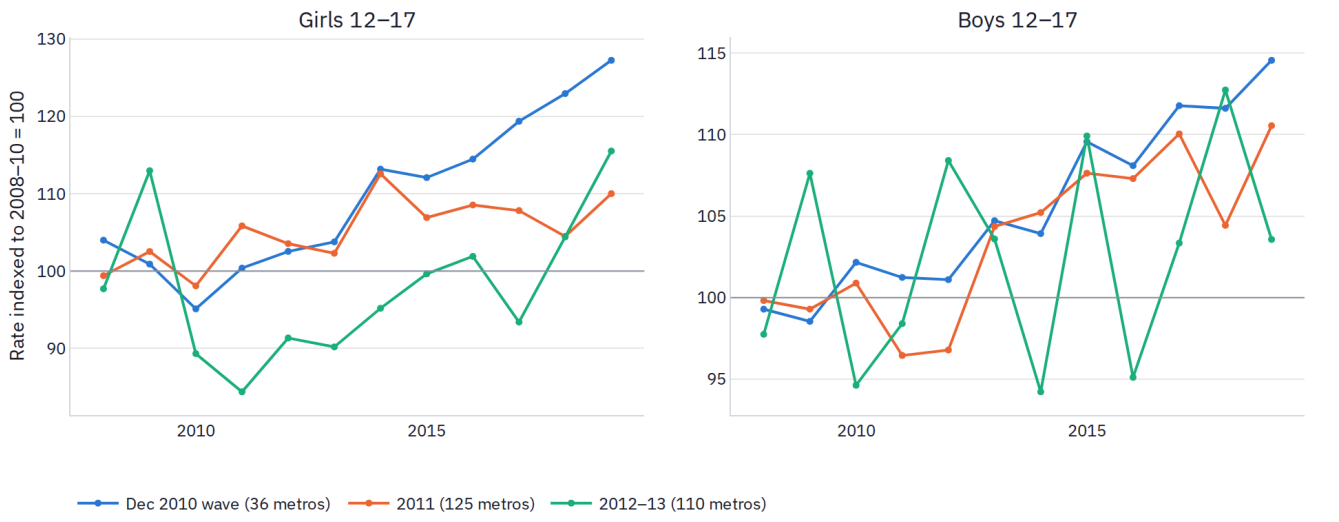
The phone hypothesis predicts that adolescent distress rose earlier in places where smartphones and fast mobile data arrived first, that heavier users experience worse mental health, and that similar patterns hold in developing countries where mobile phones are the primary way young people go online. The tests below move from coarse geographic exposure (metro 4G launch dates and state adoption shares) to individual-level exposure in low- and middle-income countries.

4G rollout across U.S. metros

If the arrival of fast mobile data drove the initial upturn in adolescent distress, teenagers in metropolitan areas that received commercial 4G LTE earliest should have experienced an earlier rise in cognitive difficulty and sadness.

We compiled commercial 4G LTE launch dates from carrier press releases and trade-press archives covering 967 carrier-market launches (Verizon 442, AT&T 243, Sprint 234, T-Mobile 47). Mapped to 2013 CBSA boundaries, these records establish a first-launch date for 98 of the 100 largest metropolitan areas and 279 of the 311 metros identified in the public ACS. Verizon's initial December 2010 rollout covered 36 of these metros, followed by 125 metros in 2011, 96 in 2012, and 14 in 2013. Each market's treatment onset is assigned to the first school year beginning after its initial commercial launch. Remaining coverage gaps (primarily Verizon's smaller 2013 rural market waves and roughly 44 AT&T launch dates outside the top 100 metros) are documented in the methods note.

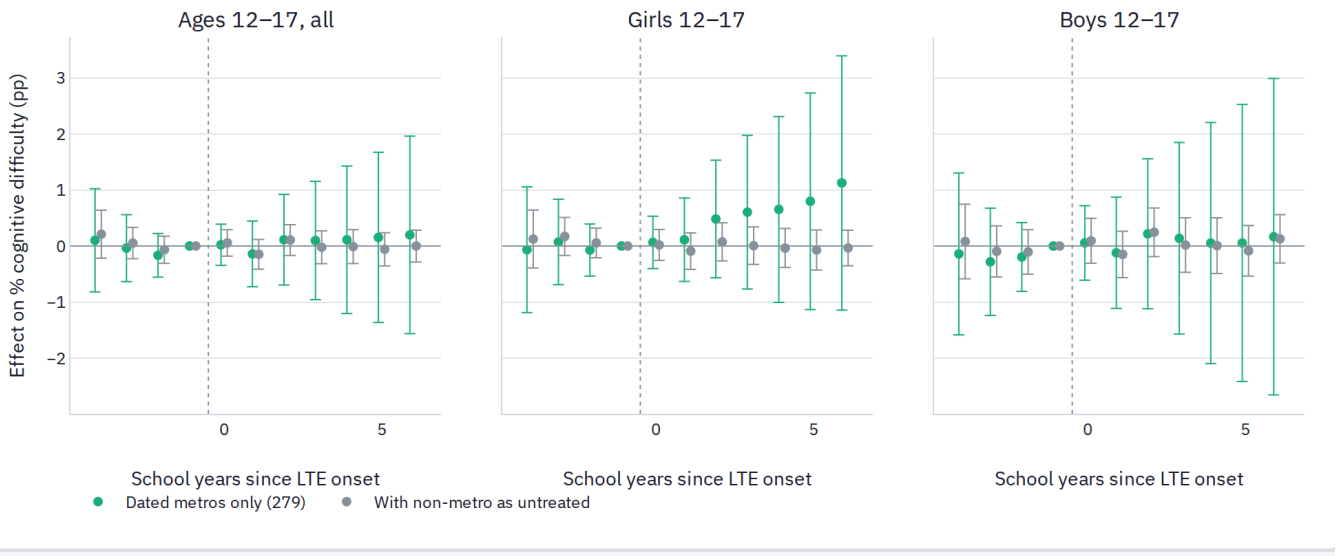
Descriptive: onset cohorts



Population-weighted ACS cognitive-difficulty rate for 12–17-year-olds in metros grouped by first commercial LTE launch, indexed to each cohort's 2008–10 mean.

If early 4G availability accelerated adolescent distress, metros in the December 2010 and 2011 launch waves should have pulled ahead of the 2012–13 wave between 2011 and 2013. Through 2013, however, the three cohorts move closely together. Starting in 2014, girls in the December 2010 cohort drift higher (reaching an index of 127 by 2019 versus 110–116 in later cohorts), but that earliest cohort consists of the 36 largest metros—where the affluent-teen composition shift documented in Part 1 is concentrated—and the divergence appears three years after launch. In the event-study regressions below, the coefficients for girls four to six years after launch are +0.6 to +1.1 pp with standard errors near 1 pp.

Event study



Estimated effect by school year relative to initial LTE launch (with the year prior to launch as the reference period), including market and year fixed effects, population weights, and standard errors clustered by market. Green: the 279 metros with a dated launch, identifying off rollout timing alone. Grey: adds the 32 undated metros and state non-metro remainders coded as never treated (an upper-bound contrast, since rural areas received LTE around 2013–14).

Onset definition	Sample	12–17 all	Girls	Boys	65+ placebo	25–49
First carrier	dated metros	+0.02 (0.15)	−0.04 (0.19)	+0.03 (0.27)	+0.03 (0.15)	+0.11 (0.08)
Verizon	dated metros	+0.01 (0.16)	−0.11 (0.20)	+0.05 (0.29)	+0.07 (0.16)	+0.11 (0.08)
Second carrier	dated metros	−0.01 (0.15)	−0.11 (0.17)	+0.20 (0.23)	+0.28 (0.14)	+0.09 (0.07)
First carrier	+ non-metro as untreated	0.00 (0.09)	−0.08 (0.10)	+0.11 (0.13)	+0.12 (0.11)	−0.10 (0.06)

Static difference-in-differences coefficient on "LTE in force" in percentage points, with standard errors in parentheses. Permuting onset dates across markets yields a placebo standard error of roughly 0.07 pp for the full panel.

Assessment. Staggered 4G LTE launch dates across U.S. metropolitan areas do not predict the timing of the rise in the ACS cognitive-difficulty item. For girls, the 95% confidence interval rules out effects larger than +0.33 pp on the dated-metro sample and +0.12 pp when including non-metro controls, compared with a national 2010–2019 increase of roughly 0.5 pp. Three limitations prevent this null result from ruling out H1: commercial LTE launches were compressed into a narrow 2011–2012 window across most major metros; adolescent social-media norms spread across markets through national platforms; and the ACS disability item captures only a fraction of internalizing distress. What the result does establish is that a metro-level rollout design using the ACS cannot detect a smartphone effect. In the next subsection we re-run the same geographic timing test on county-level youth mental-health surveys, and in the reconciliation section we compare our setup with Elaine Guo's British Columbia wireless study, which found a large effect on clinical school designations in rural mountain towns.

Rollout and adoption at county level, state youth surveys

LTE onset at county level

We link counties to 2013 CBSAs and their initial commercial 4G LTE launch dates. The unit of observation is a county-by-grade panel series with county-by-grade and state-by-year fixed effects, ensuring that changes in which grades a state surveys cannot register as changes in distress. Non-metro and undated counties serve as the comparison group (providing an upper-bound contrast, since rural LTE arrived around 2013–14), and standard errors are clustered by CBSA (or by county outside metro areas). Because the state surveys are biennial or triennial, no single pre-launch year is observed for every cohort, so event-study specifications pool the three years prior to LTE launch as the reference period.

Panel	DiD on "LTE in force" (pp)
Sad/hopeless, WA + ME counties, 2009-2019 (54 counties, 26 dated-metro, 130 county-grade series)	+0.64 (SE 0.87)
Considered suicide, WA + ME + MN counties, 1998-2019 (141 counties, 59 dated-metro, 217 county-grade series)	-0.72 (SE 0.43)
Considered suicide, MN counties, 1998-2019 (87 counties, 33 dated-metro, 87 county-grade series)	-1.82 (SE 0.67)

Smartphone adoption at the metro level

Outcome	Sex	Effect per +10 pp teen-household smartphone internet (SE)	Counties / CBSAs	States
sad hopeless	all	+0.13 (0.13)	89 / 42	CA, IL, ME, PA, WA
sad hopeless	female	+0.22 (0.25)	32 / 21	CA, ME
sad hopeless	male	-0.11 (0.14)	32 / 21	CA, ME
considered suicide	all	+0.06 (0.15)	105 / 46	CA, IL, ME, MN, PA, WA
considered suicide	female	-0.04 (0.33)	31 / 21	CA, ME
considered suicide	male	-0.39 (0.13)	31 / 21	CA, ME

NTIA CBSA smartphone-internet shares interpolated between waves (restricted to CBSAs with at least 30 unweighted respondents), with county-by-grade and state-by-year fixed effects and standard errors clustered by CBSA, 2011–2019.

Assessment. Measuring self-reported sadness and suicidal ideation at the county level yields the same conclusion as the ACS metro panel: neither 4G LTE launch timing nor metro-level smartphone adoption predicts where adolescent distress rose fastest. The only coefficient exceeding two standard errors—Minnesota suicidal ideation after LTE onset (−1.82 pp, SE 0.67)—is negative and reflects the continuation of metro counties' steeper pre-2011 decline rather than a discrete break at LTE launch. Likewise, the positive state-level YRBS correlation between smartphone adoption and girls' sadness does not carry over to the county panels (+0.13 pp per 10 points overall across five states, SE 0.13; +0.22 pp for girls in California and Maine, SE 0.25). Across counties with very different connectivity timelines, the 2010s rise occurred almost simultaneously, and the recent post-2021 waves show a broad decline.

Adoption rather than rollout: smartphone use against the state youth survey

Because commercial 4G launch dates are a coarse measure of when teenagers actually went online on phones, we next test whether state-level smartphone adoption predicts adolescent sadness. Using the harmonized NTIA series (the share of persons in households with a 12–17-year-old who use a smartphone for internet, interpolated between the 2011–2019 CPS waves) against the YRBS state files across 39 states from 2011 to 2019, with state and year fixed effects:

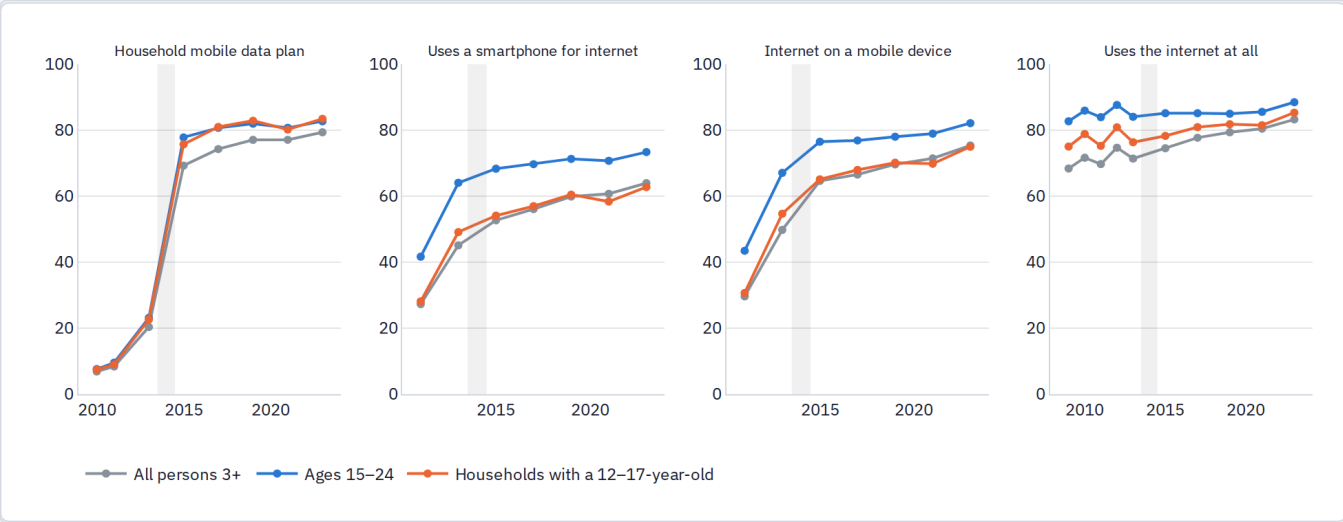
Outcome	Measure of adoption	Effect per +10 pp of adoption (SE)
YRBS sad/hopeless, girls	teen-household smartphone internet	**+0.90 (0.44)**
YRBS sad/hopeless, boys	same	+0.28 (0.41)
YRBS sad/hopeless, girls	household mobile data plan	+0.50 (0.51)
ACS cognitive difficulty 12–17, girls	teen-household smartphone internet	−0.04 (0.07)

The state-level estimates align with the pattern predicted by H1—positive for girls rather than boys, and appearing on the self-reported YRBS sadness item rather than the ACS disability screener—but the evidence is fragile: it rests on five biennial YRBS waves, interpolated state adoption shares, and a p-value near 0.04 across 39 state clusters. Moreover, the county-level version of the same regression in the state youth surveys does not reproduce this relationship, so we treat the state-level correlation as suggestive.

Smartphone adoption in the United States, 2009–2023

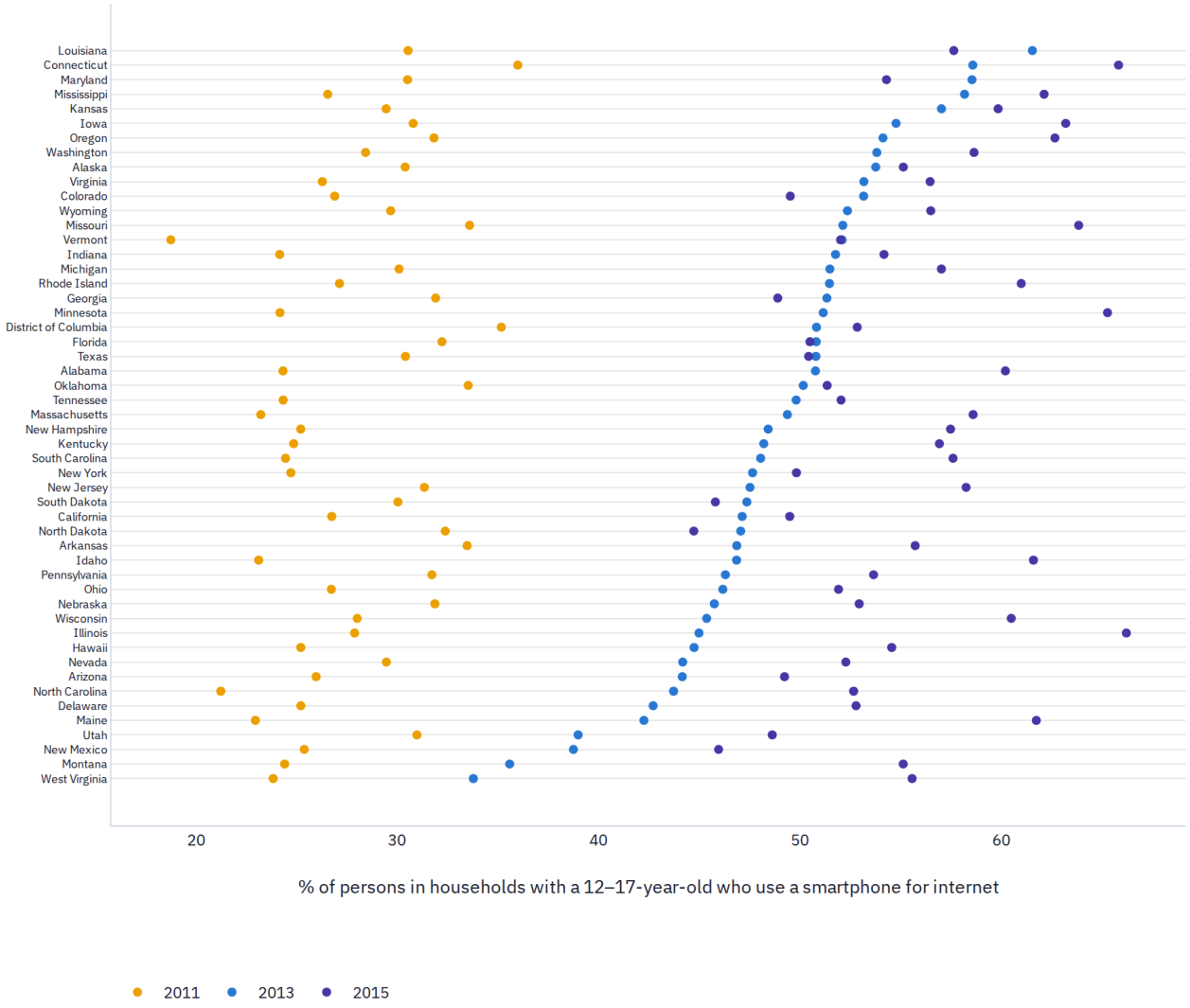
Because IPUMS CPS includes only a subset of the Computer and Internet Use Supplement and omits the mobile-phone questions, we rebuilt the adoption series from the raw Census supplement files from October 2009 through November 2023 following NTIA's harmonization code. These files supply the pre-2016 smartphone and mobile-data adoption measures that the

ACS lacks. The grey band marks the 2015 questionnaire redesign, which breaks the household mobile-data series (capturing at-home mobile broadband before 2015 and any-location mobile data plans from 2015 onward).



Weighted shares by wave. Smartphone-internet use reflects "use a cellular phone or smartphone to access the internet" in 2011 and 2013 (all persons) and NTIA's internet-enabled-phone definition from 2015 onward. In 2012 the phone items were asked only of the primary household respondent and are omitted here.

Smartphone internet use in households with a teenager rose from roughly 30% in 2011 to 45% in 2013 and 55% in 2015, before approaching saturation by 2017. Across states, teen-household smartphone use varied by 20 to 30 percentage points during the 2011–2015 transition window, providing the cross-state adoption variation used in our U.S. regression tests.

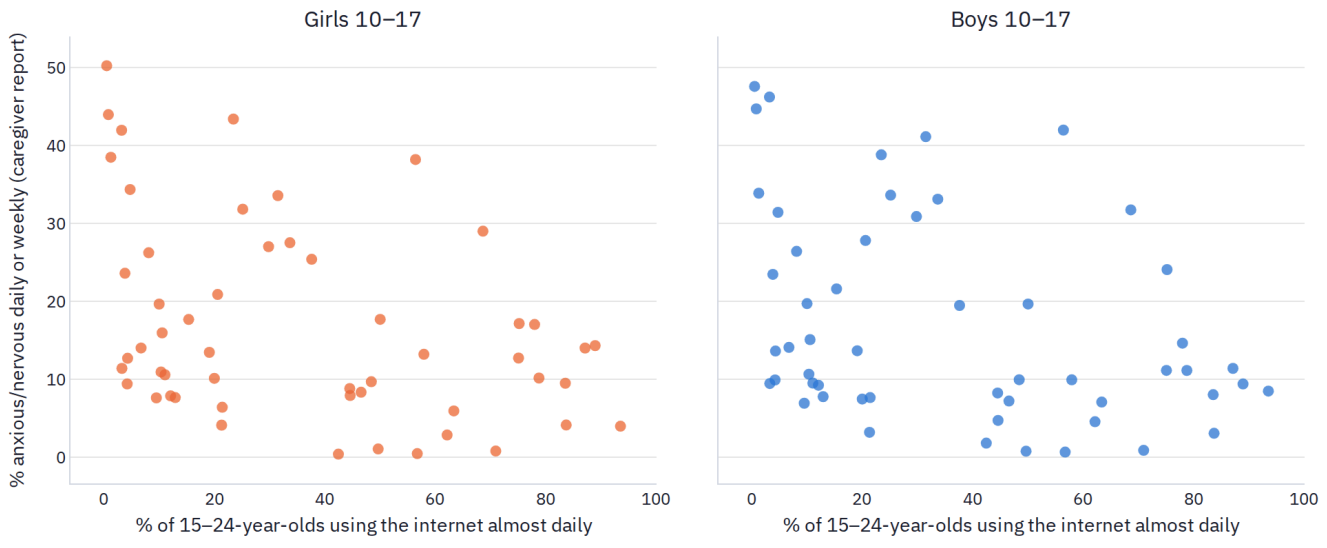


States ordered by their 2013 share. CBSA-level shares for 260–295 metropolitan areas per wave are also included in the compiled extract; county regressions restrict to CBSAs with at least 30 unweighted respondents.

Phone-first countries: MICS Child Functioning Module (68 surveys, 2017–2024)

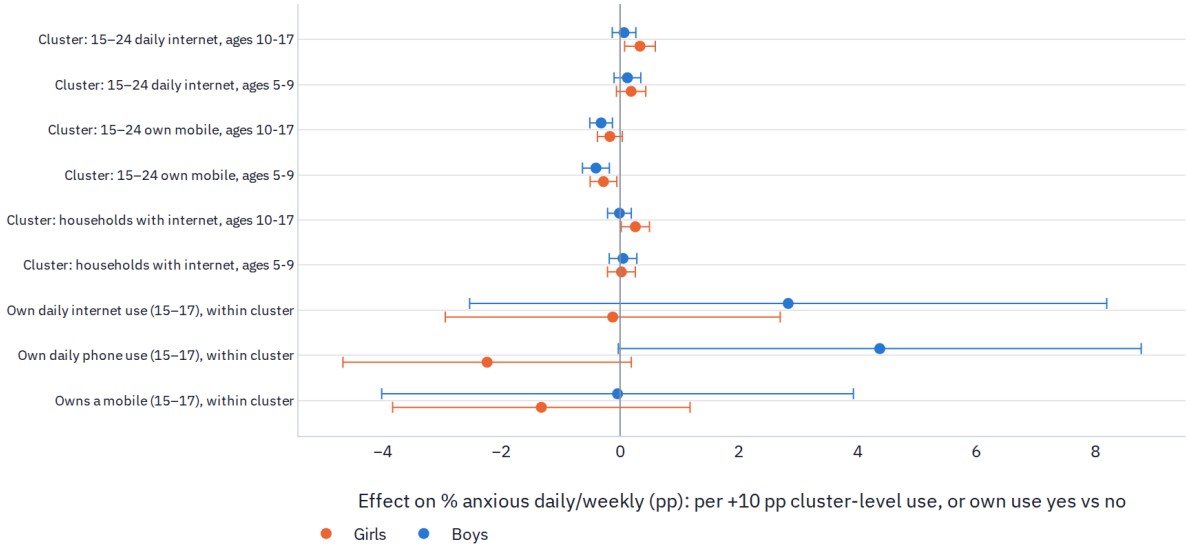
MICS6 and MICS7 surveys (2017 onward) administer the Washington Group / UNICEF Child Functioning Module for children aged 5–17, asking the mother or primary caregiver how often the child "seems very anxious, nervous or worried" and how often the child seems "very sad or depressed" (daily, weekly, monthly, a few times a year, or never), using identical wording across countries. In the same households, women aged 15–49 (and men in roughly half the surveys) report mobile-phone ownership and how often they used the internet and a mobile phone over the past three months, while the household questionnaire records home internet access. Reading all 68 MICS samples with the module on the IPUMS-MICS server yields 507,478 children with valid responses and weights across 67 surveys (53 of which also include the young-adult internet items).

Prevalence across surveys



One point per survey (ages 10–17, across the 53 surveys that also record young-adult internet use). Cross-survey variation in caregiver-reported anxiety is wide—ranging from under 1% in Laos, Tuvalu, Vietnam, and Turkmenistan to 40–50% in Tunisia, Afghanistan, the Central African Republic, and Chad—and largely reflects national reporting norms, which is why all tests below compare children within the same country and region.

Use and symptoms within countries



Cluster rows regress children's daily/weekly anxiety on the share of 15–24-year-olds in their sampling cluster who use the internet almost daily (or own a mobile phone, or the household internet share), with survey-by-region fixed effects, household wealth quintile, urban/rural status, and age controls, and standard errors clustered by sampling cluster. Own-use rows link 15–17-year-olds to their own individual record in the women's or men's questionnaire (85% of girls matched, 24% of boys) and compare users with non-users within the same sampling cluster.

Girls aged 10–17 living in sampling clusters where more young adults use the internet daily are reported as slightly more anxious (+0.34 pp per 10-percentage-point rise in the cluster share, SE 0.13). However, a similar positive association appears for 5–9-year-old girls (+0.19 pp, SE 0.13), who are too young to use phones themselves, indicating that this reflects characteristics of more connected communities rather than adolescent phone use. Cluster-level mobile-phone ownership is *negatively* associated with caregiver-reported anxiety across all age and sex groups; household internet access shows a small positive association for 10–17-year-old girls (+0.26 pp per 10 points, SE 0.12, and +0.02 pp at ages 5–9); and caregiver-reported depression shows no positive relationship with any connectivity measure. Comparing 15–17-year-olds who use the internet or a phone daily with non-users in the same sampling cluster—the closest MICS equivalent to individual-level screen exposure—yields point estimates for girls that are near zero or slightly negative: −0.1 pp on anxiety (SE 1.4) and −1.8 pp on depression (SE 1.3) for daily internet use, and −2.2 pp on anxiety (SE 1.2) for daily phone use. For boys, the within-cluster own-phone-use coefficients are positive (+4.4 pp on anxiety and +5.5 pp on depression, SE ~2.0), though these rely on the roughly one-quarter of boys interviewed in the men's sub-sample. The learning and friendship functioning items show no meaningful association in either sex.

Two-round countries

Country	Ages	Sex	Rounds	Anxious daily/weekly %	Depressed daily/weekly %
KGZ	10-14	boys	2018 → 2023	11.3 → 7.5	7.0 → 6.3
KGZ	10-14	girls	2018 → 2023	13.1 → 8.2	7.5 → 7.4
KGZ	15-17	boys	2018 → 2023	10.7 → 9.3	5.5 → 4.3
KGZ	15-17	girls	2018 → 2023	11.9 → 12.7	6.5 → 6.8
MNG	10-14	boys	2018 → 2023	7.5 → 11.2	4.4 → 7.6
MNG	10-14	girls	2018 → 2023	9.9 → 13.3	6.3 → 9.0
MNG	15-17	boys	2018 → 2023	15.0 → 11.9	7.4 → 7.5
MNG	15-17	girls	2018 → 2023	20.5 → 16.0	13.7 → 10.6
TKM	10-14	boys	2019 → 2024	0.8 → 4.7	0.7 → 2.8
TKM	10-14	girls	2019 → 2024	0.8 → 2.1	0.9 → 1.7
TKM	15-17	boys	2019 → 2024	0.8 → 4.3	0.6 → 1.3
TKM	15-17	girls	2019 → 2024	1.7 → 4.8	0.8 → 3.1
TUN	10-14	boys	2018 → 2023	42.7 → 30.1	15.8 → 13.8
TUN	10-14	girls	2018 → 2023	37.5 → 28.0	12.9 → 14.5
TUN	15-17	boys	2018 → 2023	40.7 → 34.6	15.5 → 17.5
TUN	15-17	girls	2018 → 2023	39.6 → 30.9	16.3 → 20.8

Countries with both a MICS6 and a MICS7 round on the server (Pakistan is excluded because its rounds sample different provinces). Turkmenistan rises from a very low baseline; Kyrgyzstan and Tunisia decline; Mongolia is mixed.

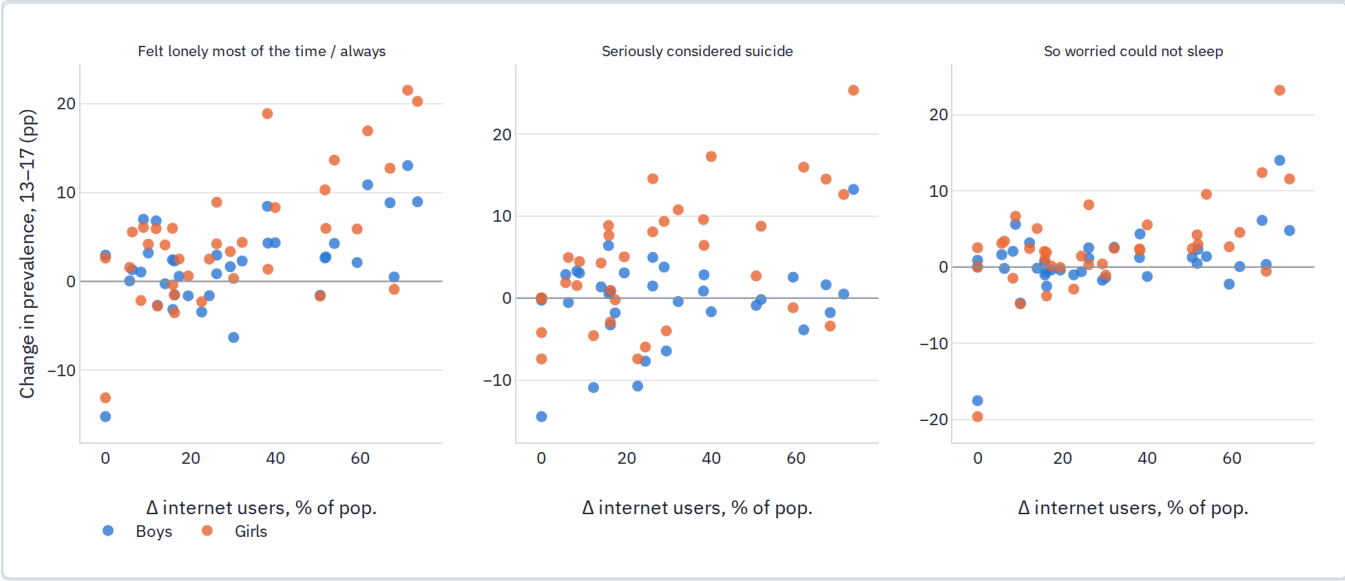
DHS-8 self-reports: Nepal 2022

The DHS-8 mental-health module provides a self-reported counterpart to the MICS caregiver reports. In 2022, Nepal administered the PHQ-9 and GAD-7 screeners to a subsample of 7,442 women aged 15–49 alongside questions on internet-use frequency and smartphone ownership (Kenya 2022 asked only whether a doctor had ever diagnosed depression or anxiety). Among Nepali women aged 15–19 (n = 1,379; 4.5% with PHQ-9 ≥ 10, 42% using the internet almost daily), daily internet users are 2.3 pp more likely to screen positive for depression (SE 2.4) and 3.5 pp more likely to screen positive for anxiety (SE 2.9) than non-users in the same sampling cluster (n = 1,293 in within-cluster regressions controlling for wealth, education, and age); the corresponding estimates are also positive at ages 20–24 and turn negative at ages 25–49. Smartphone ownership itself shows no positive association, and in Kenya diagnosed depression among 15–19-year-olds is unrelated to owning a smartphone (−0.1 pp, SE 1.3). Although these single-survey cross-sections are too small to be precise, their pattern matches the self-reported GSHS phone-hours gradient rather than the null results in the MICS caregiver reports: the positive association with screen use appears when adolescents report both their own use and their own symptoms.

Assessment. Across 67 MICS surveys, neither local young-adult connectivity nor a teenage girl's own daily internet or phone use predicts caregiver-reported anxiety or depression within sampling clusters, and the small positive cluster correlation for girls' anxiety is shared by 5–9-year-olds. Three qualifications apply: the outcome is reported by caregivers rather than adolescents themselves; the exposure measures current use rather than exogenous network rollout timing; and only four countries currently offer repeated MICS rounds with the module.

Phone-first countries: GSHS school surveys (149 country-rounds, 2003–2023)

The WHO Global School-based Student Health Survey (GSHS) asks school-attending adolescents (primarily ages 13–17) in over 100 low- and middle-income countries identical questions on loneliness, anxiety-related sleep loss, and suicidality in waves starting in 2003. From the 165 public-use country-round files in the WHO microdata repository, 149 rounds contained usable mental-health items and survey weights after harmonization, including 91 rounds across 39 countries surveyed at least twice. We first test whether within-country changes in adolescent distress track national internet penetration from the World Bank (controlling for log GDP per capita), and then examine student-level phone hours and ITU mobile-network coverage below.



Each point plots a country's change from its first to its last GSHS wave (weighted prevalence) against the change in the national internet-user share over the same interval.

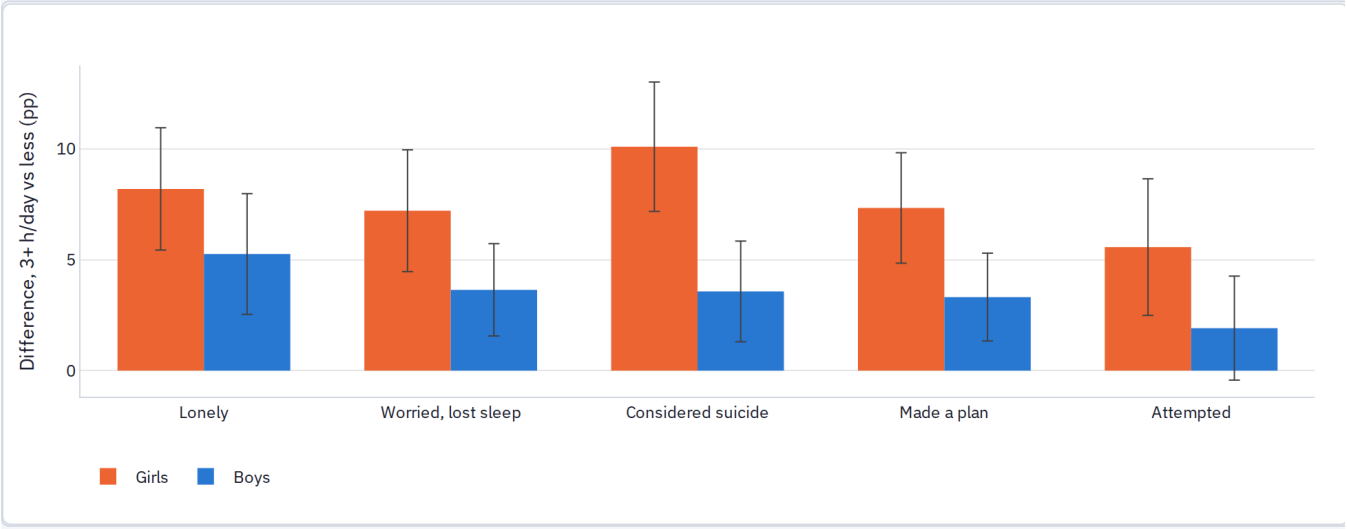
Outcome	Sex	Sample	Effect of +10 pp internet users (SE)
lonely	girls	79 rounds / 34 countries	+1.04 (0.83)
lonely	boys	79 rounds / 34 countries	-0.18 (0.59)
worried sleep	girls	77 rounds / 33 countries	+0.14 (0.87)
worried sleep	boys	77 rounds / 33 countries	-0.78 (0.54)
suicide consider	girls	67 rounds / 30 countries	+0.02 (1.27)
suicide consider	boys	67 rounds / 30 countries	-0.42 (0.99)
suicide plan	girls	70 rounds / 31 countries	+0.86 (0.74)
suicide plan	boys	70 rounds / 31 countries	-0.35 (0.69)
suicide attempt	girls	35 rounds / 19 countries	+0.30 (0.90)
suicide attempt	boys	35 rounds / 19 countries	-0.50 (0.38)
no close friends	girls	79 rounds / 34 countries	+0.48 (0.33)
no close friends	boys	79 rounds / 34 countries	-0.16 (0.36)

Country and year fixed effects, log GDP per capita control, inverse-variance weights, and standard errors clustered by country.

Assessment. Across the full 34-country panel, within-country growth in national internet penetration shows no statistically significant relationship with adolescent distress: for girls, a 10-percentage-point increase in internet users is associated with +1.04 pp in loneliness (SE 0.83) and +0.02 pp in suicidal ideation (SE 1.27), while the point estimates for boys are slightly negative. (An earlier partial extract of 72 rounds had suggested a +2.5 pp effect for girls on both outcomes, which disappeared once the remaining 77 rounds were added—a useful illustration of how sensitive country-round panels are to sample coverage.) National internet penetration is also a coarse proxy that pools adult and youth access. The two subsections below turn to the 2022–23 GSHS waves that ask students directly about daily mobile-phone hours and to the ITU mobile-broadband coverage series.

Asking adolescents directly: mobile-phone hours in the 2022–23 rounds

Five recent GSHS rounds (Ghana 2022, India 2022, Jamaica 2023, Morocco 2023, and Wallis and Futuna 2023) added a question on "hours per day you used a mobile phone in the past 7 days," pairing each student's self-reported phone time with the same student's loneliness and suicidality responses.



Regressions include country fixed effects and age controls, survey weights, and standard errors clustered by school across 7,708 girls and 7,102 boys (including students without a phone). Compared with students using a phone under three hours a day or not at all, girls at three or more hours report loneliness +8.2 pp (SE 1.4), suicidal ideation +10.1 pp (SE 1.5), and suicide attempts +5.6 pp (SE 1.6); the corresponding differences for boys are +5.3, +3.6, and +1.9 pp. Distress rises steadily across higher-use categories: relative to under one hour per day, girls reporting more than eight hours per day have +19.5 pp higher loneliness, +17.4 pp higher suicidal ideation, and +13.7 pp higher suicide attempts. Students without a mobile phone sit a few percentage points above the under-one-hour group rather than below it.

This individual-level dose-response gradient matches the pattern documented in Western surveys: substantial in magnitude, roughly twice as steep for girls as for boys, and increasing monotonically with daily hours above one hour. Because it is a cross-sectional self-report association, however, it cannot distinguish the effect of phone use on distress from reverse causation or shared underlying vulnerabilities. What it does show is that the 2022–23 GSHS rounds (alongside incoming MICS7 adolescent modules) are strong candidates for linking to subnational mobile-coverage rollout maps, since exposure is measured directly on the adolescent rather than inferred from national or adult averages.

Network coverage at country level (ITU)

Replacing World Bank internet penetration with the ITU mobile-network series allows us to test country-level 3G and 4G population coverage and active mobile-broadband subscriptions using the same two-way fixed-effects specification (country and year fixed effects, inverse-variance weights, and country-clustered standard errors across countries with two or more GSHS rounds).

Exposure (per +10 points)	Countries / rounds (loneliness)	Lonely, girls	Lonely, boys	Considered suicide, girls	…boys	Attempted, girls	…boys
Mobile-broadband subs per 100 (ITU)	22 / 47	+0.21 (0.54)	+0.03 (0.42)	+0.28 (0.43)	+0.35 (0.48)	+0.31 (0.44)	+0.30 (0.50)
3G population coverage (ITU)	17 / 36	-0.72 (0.28)	+0.25 (0.26)	+0.10 (0.59)	+0.28 (0.44)	-0.08 (0.32)	+0.36 (0.36)
4G population coverage (ITU)	15 / 32	-0.40 (0.41)	-0.48 (0.38)	-0.51 (0.63)	-0.20 (0.59)	-0.64 (0.43)	-0.45 (0.69)
Internet users % (ITU)	33 / 78	+0.93 (0.86)	-0.19 (0.60)	-0.04 (1.20)	-0.54 (1.07)	+0.37 (1.02)	-0.22 (0.86)

Sample counts in the second column refer to the loneliness item; suicidal ideation and suicide attempts are available in 1 to 3 fewer countries per row.

None of the country-level ITU series supports H1. Across the 15 countries with repeated 4G data, a 10-percentage-point increase in 4G coverage has slightly negative point estimates across all outcomes (from −0.20 to −0.64 pp, with standard errors of 0.38–0.69). For 3G coverage (17 countries), the coefficient on girls' loneliness is negative (−0.72 pp, SE 0.28) and the suicidality estimates are near zero; for active mobile-broadband subscriptions (22 countries), estimates range from +0.03 to +0.35 pp with standard errors around 0.4–0.5 pp. Because national coverage series move with broader economic development and pool urban and rural areas together, local cluster-level coverage remains the necessary test.

Phone-first countries: census microdata (IPUMS International)

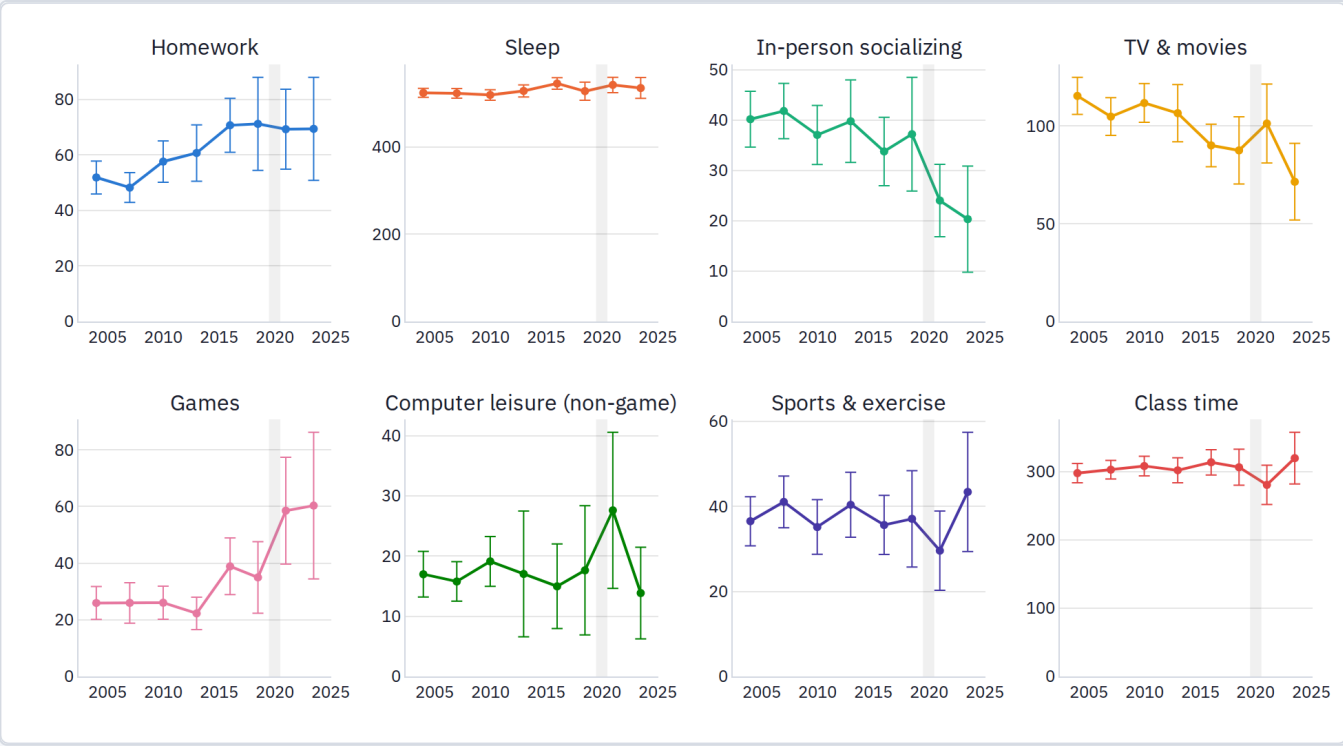
We also examined the 2009/2019 Kenya and 2010/2020 Mexico census pairs in IPUMS International, the two best candidates carrying disability items alongside connectivity measures. Both 2020-round censuses include the Washington Group question on difficulty remembering or concentrating, and the 2019 Kenya census asks each person directly about mobile-phone ownership and internet use. However, the 2010-round censuses asked narrow yes/no questions about "mental disability," so changes between the 2010 and 2020 rounds are dominated by questionnaire wording. Within the 2019/2020 cross-sections, cognitive-difficulty prevalence at ages 12–17 is low (0.8–1.8% for any difficulty and 0.2–0.7% for "a lot of difficulty" or "cannot do at all"), boys exceed girls at every age, and subnational regressions of the 12–17 rate on adolescent phone ownership or internet use across 76 Kenyan district-sex cells or on household internet across 3,259 Mexican municipality-sex cells yield coefficients within ±0.01 pp per percentage point of phone ownership or internet access. In developing-country censuses, this item behaves like a measure of severe developmental disability and is insensitive to adolescent internalizing distress.

Part 3 · School pressure (H2)

The school-pressure hypothesis predicts that adolescent distress rose when and where academic demands intensified: during the high-stakes high-school years, in states that introduced test-based accountability reforms, and in countries where students report rising schoolwork pressure. We test this channel using U.S. adolescent time-use diaries, the cross-national HBSC schoolwork-pressure series, and state accountability policy timelines.

Did school demands rise? Time use of 15–17-year-olds (ATUS, 2003–2024)

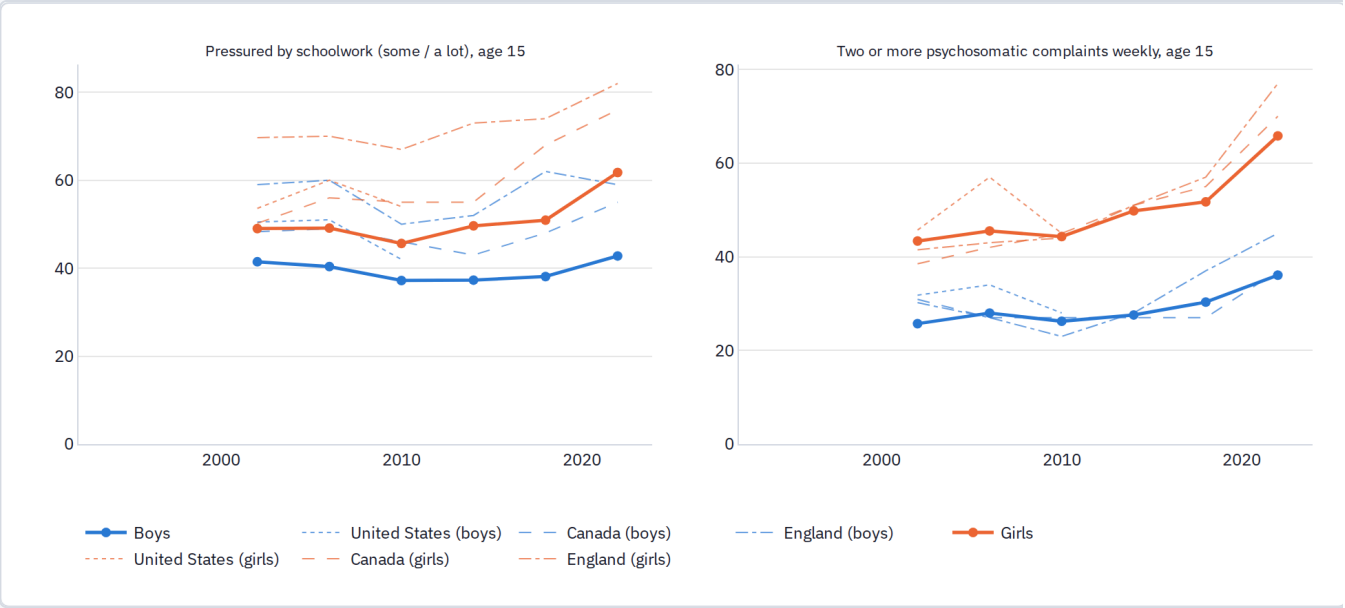
The American Time Use Survey (ATUS) measures how U.S. 15–17-year-olds allocate their time on school-year weekdays. Between 2003–08 and 2015–19, average weekday homework time rose from roughly 50 minutes to 71 minutes per day (+40%) and remained near 69 minutes through 2023–24. Over the same period, sleep duration did not decline (averaging 8.7–9.1 hours), in-person socializing drifted downward before dropping sharply after 2020, television viewing fell, and time spent playing computer or video games more than doubled. Sample sizes are modest (roughly 100 to 800 teens per pooled period), and because ATUS records only primary activities, background smartphone checking during homework or socializing is not captured.



Weighted minutes per day on non-holiday weekdays during the school year (September–May), pooled periods; vertical bars show 95% confidence intervals.

Schoolwork pressure across countries (HBSC)

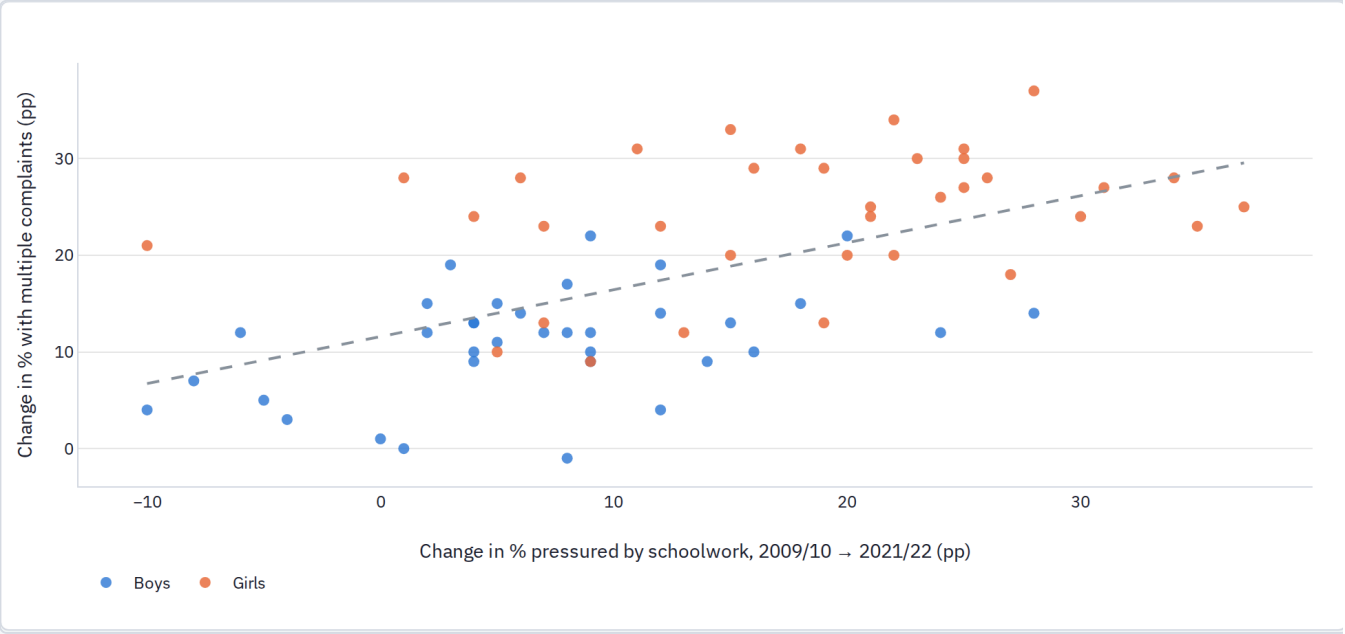
The WHO Health Behaviour in School-aged Children (HBSC) surveys ask 11-, 13-, and 15-year-olds across roughly 45 European and North American countries the same question on schoolwork pressure and the same psychosomatic-complaints checklist every four years. Published country tables from 1997/98 through 2021/22 (supplemented by the HBSC data browser for 2014–2022) allow a direct cross-national test of Gray's school-pressure mechanism. Because the United States participated only through the 2009/10 wave, post-2010 U.S. trends cannot be tracked in HBSC.



Unweighted means across participating countries (thick lines), with the United States, Canada, and England shown as thin dashed lines. Age 15.

Both series were flat or declining through 2009/10 and then rose in tandem: among 15-year-old girls, the cross-country mean of perceived schoolwork pressure increased from 46% in 2010 to 50% in 2014, 51% in 2018, and 62% in 2022, while the share reporting two or more weekly psychosomatic complaints rose from 44% to 50%, 52%, and 66%. In the 2000s, American 15-year-olds reported above-average schoolwork pressure, though U.S. pressure fell between 2006 and 2010, the final U.S. wave.

Did complaints rise where pressure rose?



Each point is a country-sex cell at age 15; the dashed line is the pooled OLS fit (slope 0.24, SE 0.07, r = 0.59). Over the pre-pandemic 2010 → 2018 window alone, the slope is 0.16 (SE 0.07, r = 0.31).

Specification (ages 11–15 pooled unless noted)	Estimate	Interpretation
Within-country panel regression of complaints on pressure, with country×age×sex and wave×age×sex FE (1,440 cells)	0.14 (SE 0.04); girls 0.16, boys 0.12	A 10-pp rise in pressure corresponds to a 1.4-pp rise in complaints
Long difference 2010 → 2018, age 15 (72 country-sex pairs)	mean Δpressure +5.5 pp, Δcomplaints +6.6 pp; slope 0.16	Cross-country pressure differences account for ~1 pp of the 6.6-pp rise
Long difference 2010 → 2022, age 15 (70 pairs)	Δpressure +12.7, Δcomplaints +17.7; slope 0.24, r = 0.59	Stronger relationship, though 2022 includes pandemic disruptions
Pre-smartphone baseline period 2002 → 2010	Δpressure −2.9, Δcomplaints −0.1; slope 0.14	Same within-country slope during a period when neither series was rising
2018 & 2022 cross-country: complaints on problematic social media use (PSMU) and pressure, wave×age×sex FE (496 cells)	PSMU 0.53 (SE 0.15); pressure 0.13 (SE 0.06)	Both predictors are independently associated with complaints
Within-country change 2018 → 2022 on PSMU and pressure	PSMU 0.25 (SE 0.18); pressure 0.12 (SE 0.10)	Imprecisely estimated with a single four-year change per country

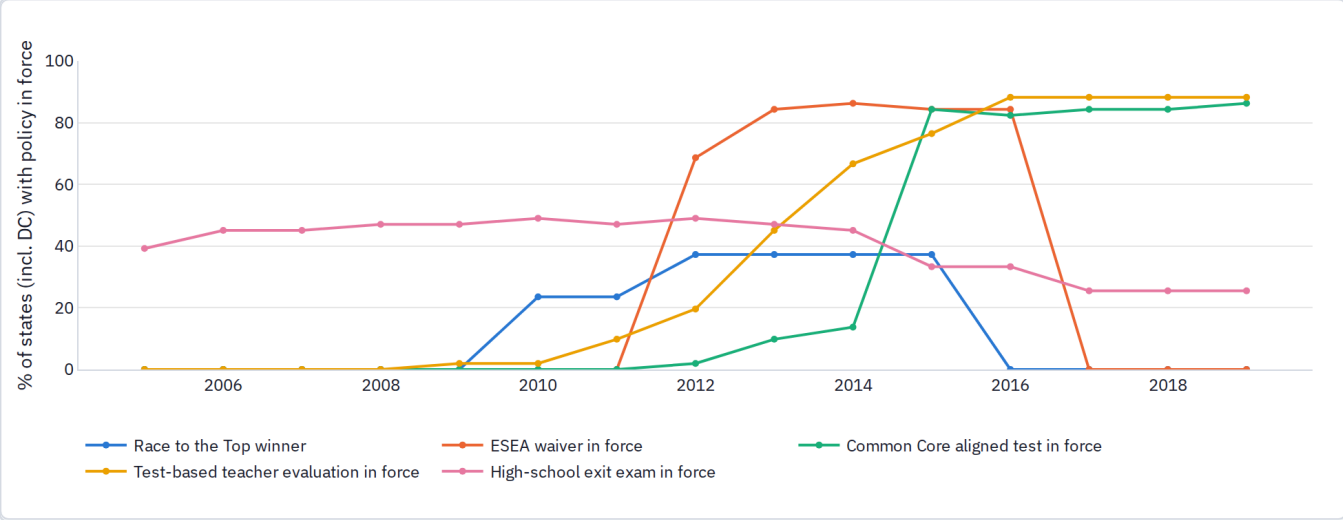
Assessment. Perceived schoolwork pressure rose after 2010 across most HBSC countries, and psychosomatic complaints rose more in countries where pressure increased more. However, the within-country slope is modest (0.14) and identical to the slope during the 2002–2010 baseline period when neither measure was rising. Perceived pressure thus behaves like a steady correlate of complaints rather than the sole driver of the post-2010 inflection, most of which is absorbed by common wave effects across countries. Testing whether complaints became more concentrated among high-pressure students within countries requires the student-level HBSC microdata.

U.S. school accountability policy

In the United States, states adopted test-based accountability reforms on staggered schedules between 2009 and 2017; if those policy shifts drove adolescent distress, states implementing them should show corresponding increases in youth depression and sadness.

We compiled implementation dates across all 51 jurisdictions for five major state accountability policies, documenting a primary source for each cell: Race to the Top awards (2010–11), ESEA/NCLB flexibility waivers (2012–16), the first operational Common Core–aligned state assessment (2012–2019, concentrated in spring 2015), statewide test-based teacher evaluation

(2009–2017), and mandatory high-school exit exams (which declined from 25 states to 13 over the decade). Years refer to the school year ending that spring.



Share of states (including DC) with each policy in force. Most states adopted ESEA waivers, teacher-evaluation rules, and Common Core–aligned tests within a narrow 2012–2015 window, leaving limited timing variation across states.

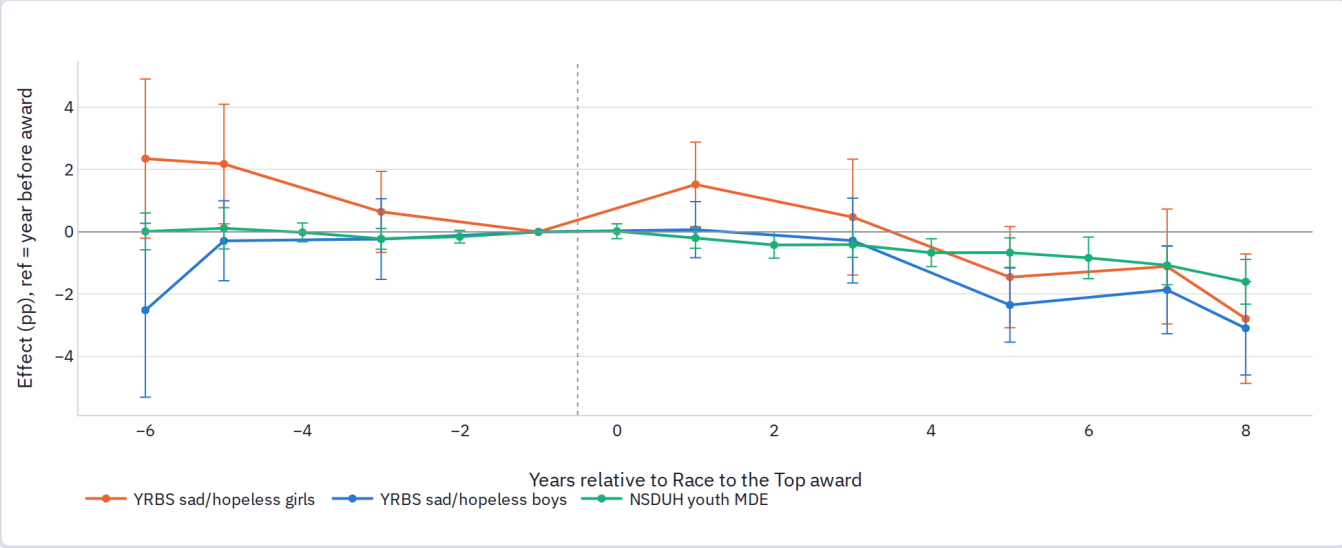
State-year regressions

Policy (each estimated alone)	ACS girls	ACS boys	YRBS sad girls	YRBS sad boys	NSDUH youth MDE
Race to the Top winner (award year–2015)	-0.06 (0.07)	-0.00 (0.10)	**+1.21 (0.41)**	**+0.68 (0.34)**	+0.16 (0.14)
ESEA waiver in force	-0.01 (0.08)	+0.13 (0.07)	-0.99 (0.58)	-0.16 (0.48)	-0.18 (0.25)
Common Core aligned test in force	-0.00 (0.08)	+0.01 (0.12)	-0.80 (0.79)	-0.80 (0.77)	-0.35 (0.30)
Test-based teacher evaluation in force	-0.03 (0.06)	-0.13 (0.09)	-0.38 (0.60)	-0.28 (0.36)	-0.03 (0.17)
High-school exit exam in force	-0.04 (0.07)	-0.03 (0.12)	-0.47 (0.82)	-0.09 (0.86)	+0.34 (0.27)
Pressure index (sum of the five)	-0.03 (0.03)	-0.01 (0.04)	-0.11 (0.25)	-0.04 (0.20)	+0.02 (0.09)

Coefficients in percentage points from regressions with state and year fixed effects, inverse-variance weights, and standard errors clustered by state, 2008–2019 (YRBS biennial across 39 states). NSDUH estimates are reported in two-year waves and matched to the average policy indicator across the two survey years; state clustering accounts for the half-sample overlap between consecutive waves (restricting to non-overlapping waves changes every NSDUH coefficient by less than 0.15 pp). Bold = |t| > 1.96.

Common Core–aligned testing, ESEA waivers, test-based teacher evaluations, high-school exit exams, and the five-policy sum show no positive relationship with any of the three distress measures; several point estimates are slightly negative. The only statistically significant positive coefficient in the table is for Race to the Top on YRBS sadness (+1.21 pp for girls, SE 0.41; +0.68 pp for boys, SE 0.34), while the corresponding NSDUH estimate is +0.16 pp (SE 0.14) and the ACS coefficients are near zero. However, the 19 Race to the Top winners were a selected group of states (including DC, New York, Massachusetts, and Florida), and the static indicator was coded as 1 only during the 2010–2015 grant window—treating the winners' higher post-2015 sadness levels as part of the control baseline. When we trace the winning states year by year around the award date in the event study below, the apparent effect disappears.

Race to the Top, traced year by year



State and year fixed effects, plotting coefficients relative to the year before the award (the school year ending spring 2010 for Phase 1 and 2 winners, and spring 2012 for the December 2011 Phase 3 winners), 2005–2019. Both award cohorts are observed in the YRBS wave prior to their award (2009 and 2011); event years 6 or more before and 8 or more after are pooled into endpoint bins.

Tracing Race to the Top winners year by year around the grant award shows why the static regression coefficient in the previous section was misleading: after winning an award, states' YRBS sadness and NSDUH youth depression rates actually trended flat to downward relative to non-winning states. The positive static coefficient was an artifact of coding the grant indicator as 1 only during 2010–2015, which placed the winning states' post-2015 observations into the reference group. The event study provides no evidence that Race to the Top increased adolescent distress. Within the U.S. public data, state accountability policy shocks do not explain the post-2010 rise.

Part 4 · The interaction (H3): phones in the same hours as high-pressure school

In its sharpest form, the interaction hypothesis (H3) proposes that smartphones harm adolescents primarily when they pervade the school day alongside high-stakes academic pressure—layering constant social comparison and notifications onto the same hours that carry grades and testing. That mechanism implies three testable predictions: academic demands and smartphone use should both have risen after 2010 (as seen in HBSC schoolwork pressure and U.S. weekday homework time); removing phones from the school day should improve well-being, particularly for students experiencing high academic pressure; and screen use outside high-pressure school environments should be less strongly tied to distress.

What the phone-ban evidence says about this

PISA 2022 allows an initial test of whether school phone bans matter more for academically anxious students. Because PISA 2022 did not include general schoolwork-anxiety items, we use the PISA mathematics-anxiety index (asked of all students) as a proxy for academic pressure, splitting students at their country's median. We estimate the same within-country regression as above, interacting the school ban indicator with high mathematics anxiety separately by sex.

Outcome	Sex	Ban, low-anxiety students (SE)	Ban, high-anxiety students	Difference (SE)	High vs low anxiety, no ban
Lonely at school	girls	-0.27 (0.68)	-1.15	-0.88 (0.93)	+8.2
Lonely at school	boys	+1.12 (0.53)	-0.81	-1.94 (0.98)	+7.4
Nervous without device	girls	-0.75 (0.60)	-0.49	+0.26 (0.78)	+9.3
Nervous without device	boys	+0.13 (0.53)	+0.19	+0.06 (0.81)	+5.8
Belonging index	girls	+2.34 (1.60)	+2.46	+0.11 (1.90)	-22.7
Belonging index	boys	+0.32 (1.43)	+1.91	+1.59 (2.11)	-19.2
Life satisfaction index	girls	+3.09 (3.41)	+2.23	-0.86 (4.53)	-25.3
Life satisfaction index	boys	+2.54 (3.26)	+0.71	-1.82 (4.93)	-25.2

Binary items in percentage points and scale indices in hundredths of an SD; regressions include country fixed effects, socio-economic status, private/public school type, and grade controls, student weights, and school-clustered standard errors across 441,992 students with non-missing mathematics anxiety.

Students with above-median mathematics anxiety are substantially lonelier at school (+8.2 pp among girls and +7.4 pp among boys in non-ban schools), and the association between attending a ban school and lower loneliness is larger for them: −1.15 pp for high-anxiety girls versus −0.27 pp for low-anxiety girls (interaction −0.88 pp, SE 0.93), and −0.81 pp versus +1.12 pp for boys (interaction −1.94 pp, SE 0.98). These interaction estimates point in the direction H3 predicts, though they are noisy and rely on mathematics anxiety as a proxy for broader academic pressure. Notably, school bans reduce daily in-school phone use by virtually the same amount (~13 percentage points) in both anxiety groups, so any difference in well-being would reflect differential sensitivity to removing phones rather than differential compliance. A more direct test of H3 requires pairing school phone policies with a direct measure of schoolwork pressure, which the 2021/22 HBSC microdata can support.

Did 4G timing matter more in high-pressure states?

Outcome (ACS 12–17, market panel)	LTE × policy	Interaction (SE)
girls	pressure_index	-0.34 (0.25)
girls	cc_aligned_test_in_force	not identified
girls	waiver_in_force	+0.12 (1.02)
girls	rttt_winner	-0.94 (0.49)
boys	pressure_index	+0.36 (0.34)
boys	cc_aligned_test_in_force	not identified
boys	waiver_in_force	-0.04 (1.63)
boys	rttt_winner	+0.61 (0.75)

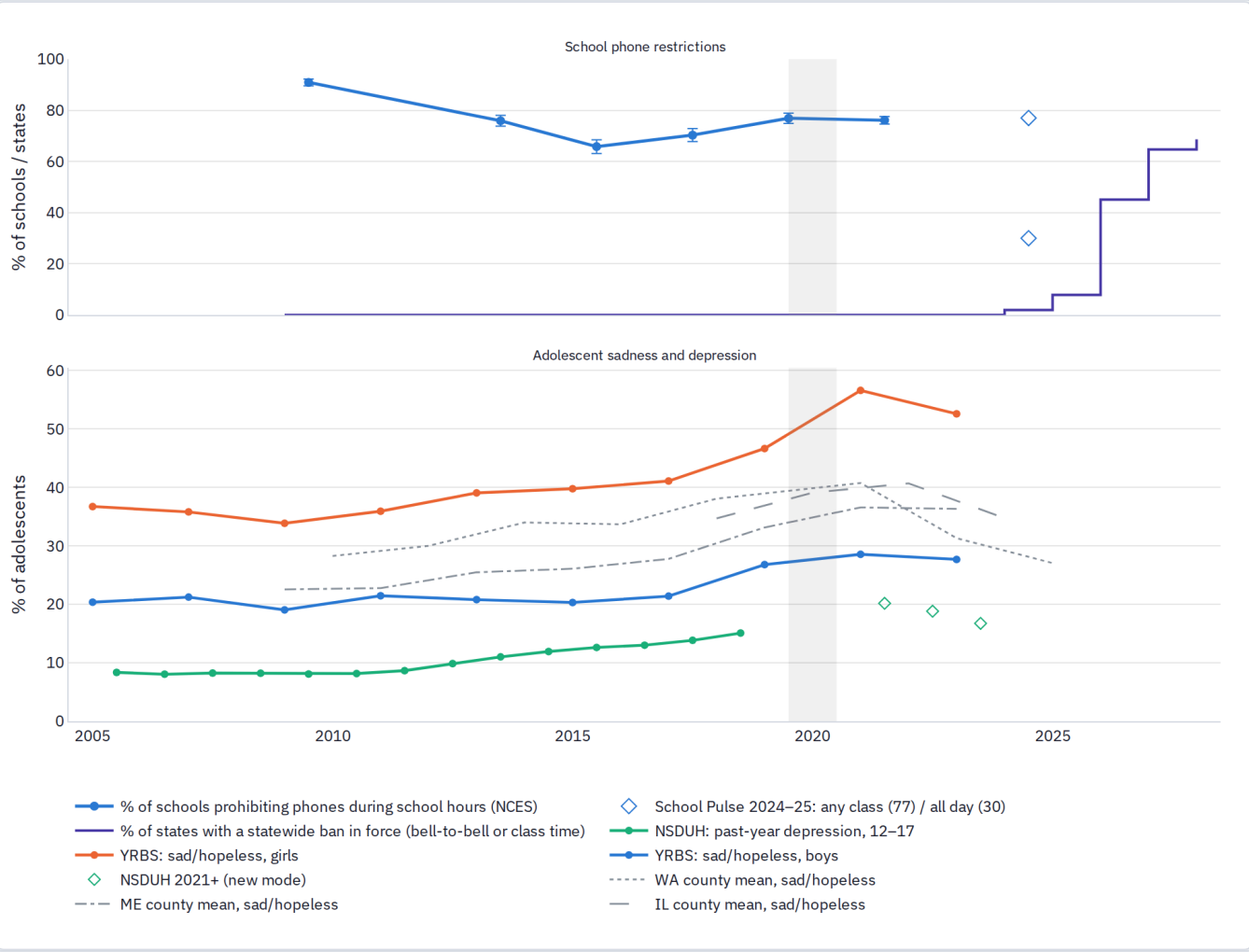
State-by-year fixed effects absorb the direct effect of state policies; the interaction tests whether early-LTE metros rose more than late-LTE metros within states that had each policy in force. "Not identified" indicates that the policy turned on in the same year as LTE for nearly all dated metros in the state.

Where the interaction is identified, the point estimates for girls are negative (LTE × pressure index −0.34 pp, SE 0.25; LTE × Race to the Top −0.94 pp, SE 0.49). Early-LTE markets inside high-accountability states did not experience a larger rise in the ACS cognitive-difficulty item, though this test carries the same metro-level limitations as the main LTE event study.

School phone bans

School phone restrictions are the one policy intervention that both the phone-centered and interaction hypotheses predict should matter, and they have shifted markedly over the past fifteen years. Below we combine national school-policy surveys, our 51-state legislative tracker, existing quasi-experimental evaluations, and PISA 2022 microdata to assess what school phone bans do.

Bans and sadness on one time axis



Top: principal-reported share of U.S. public schools prohibiting non-academic phone use during school hours (NCES SSOCS; 2013–14 is from a smaller Fast Response Survey; question wording changed in 2017–18), the 2024–25 School Pulse Panel estimates (77% prohibiting use during any class, 30% all day), and the cumulative share of states with a statewide bell-to-bell or instructional-time mandate in force. Bottom: NSDUH past-year major depressive episode (ages 12–17), YRBS persistent sadness by sex, and county-mean sadness from the Washington, Maine, and Illinois youth surveys. Grey band = 2020.

Plotted on a common timeline, U.S. school phone restrictions and adolescent distress moved in opposite directions over two successive cycles. Between 2009–10 and 2015–16—the years when smartphones replaced feature phones—the share of public schools prohibiting cell-phone use fell from 91% to 66%, while youth depression and sadness rose from their 2011–13 lows. School prohibitions then rebounded to 77% by 2019–20, and a wave of state legislation between 2023 and 2026 has pushed mandatory class-time or bell-to-bell bans into a majority of states; over the same post-2021 window, adolescent sadness fell across every state survey with recent data. Because these are aggregate national series, however, the timing alignment is only suggestive: sadness continued to rise through 2019–2021 even after school prohibitions began recovering, and the

post-2021 downturn started before most state laws took effect. Two non-exclusive mechanisms could account for the recent decline:

- **Gradual school re-restriction and broader norm shifts.** The recovery in school restrictions after 2016, combined with growing concern among parents, educators, and teenagers themselves, may have curbed the most disruptive in-school phone use even before statutory state mandates arrived. That interpretation is consistent with the PISA 2022 cross-section and Norway's middle-school ban study, though the aggregate U.S. timing is loose.
- **Shifts in platform content.** If the 2012–2019 rise was driven primarily by peer social comparison on photo-sharing networks (Instagram, Snapchat), the subsequent shift of teen screen time toward algorithmic short-form video feeds (TikTok, YouTube) could keep total screen hours high while generating less acute social-comparison distress. Because public surveys do not track platform mix over time, our data cannot test this channel directly.

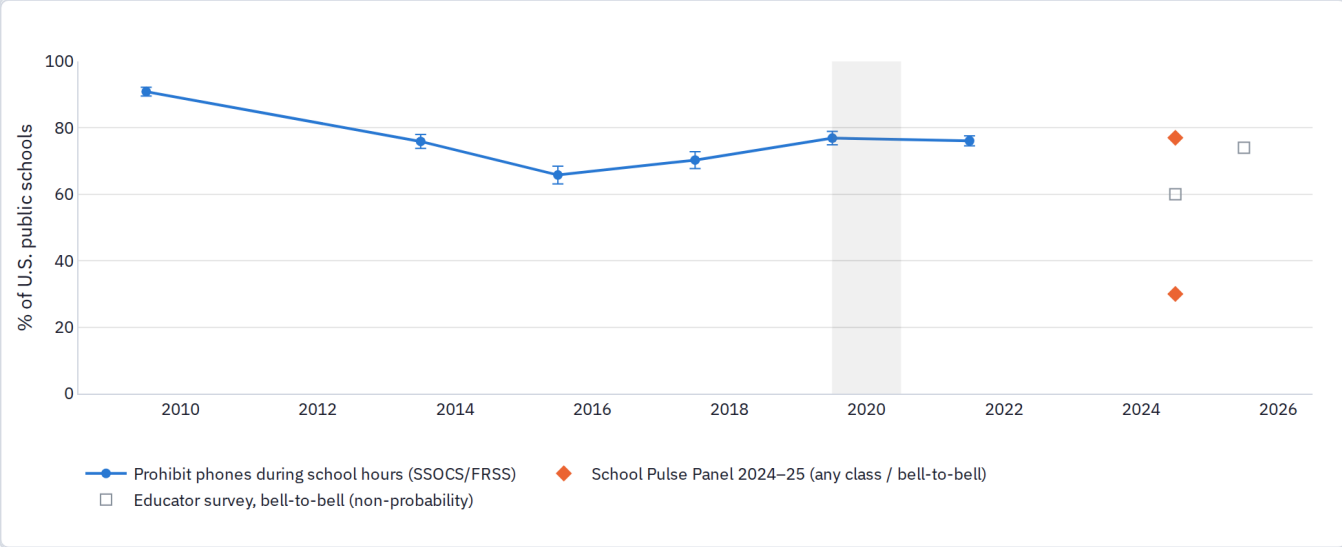
Lags and enforcement

Two practical features of school phone bans also affect how the aggregate timeline and upcoming state evaluations should be interpreted:

- **Effects take a year or two to materialize.** Quasi-experimental studies consistently find adjustment lags after a ban is introduced. In the 2026 U.S. Yondr-pouch study (Allcott et al.), student subjective well-being *dipped* by 0.20 SD in the adoption year before turning positive (+0.16 SD) in year two; in Norway (Abrahamsson 2024), reductions in bullying and mental-health visits grew over three years; and in Florida (Figlio and Özek 2025), student suspensions rose 12% in year one before fading in year two as test-score gains emerged. If school bans require a year of administrative adjustment before improving school climate, the 2015–16 trough in U.S. restrictions would register around 2017–18 and the 2024–25 state mandates around 2026–27. Evaluations of the new U.S. laws will therefore need at least two post-ban waves rather than a single immediate post-year.
- **Stated policy differs from actual enforcement.** PISA 2022 measures both principal-reported school bans and student-reported daily smartphone use at school. Across 52 education systems with both items, a median 46% of students in ban schools still reported using a smartphone at school daily (versus 62% in non-ban schools), and in 24 systems a majority of students in ban schools did so. Enforcement in the United States in 2022 was particularly weak: 70.1% of students in U.S. ban schools used a smartphone daily at school compared with 81.7% in non-ban schools (an 11.6-point gap, versus gaps of 30–46 points in Japan, Taiwan, Switzerland, and Brunei). Stated school policies in the pre-2024 U.S. data were thus loosely enforced, and any effects estimated from early state mandates will represent a lower bound on strict bell-to-bell enforcement.

Country-by-country daily in-school smartphone rates in ban and non-ban schools are reported in `results/tables/pisa_ban_enforcement_by_country.csv`.

The national school-policy series



Principal-reported share of public schools prohibiting non-academic cell-phone use during school hours (NCES SSOCS; the 2013–14 point comes from a smaller Fast Response Survey, and the item wording changed in 2017–18). The 2024–25 School Pulse Panel asks about class time rather than all school hours: 77% prohibit phones in any class, and 30% prohibit them both in class and outside class.

State bans and what can be tested when

Policy type	Effective school year	N	States
ban during instructional time	2024-25	1	Indiana
ban during instructional time	2025-26	5	Iowa, Nevada, North Carolina, Tennessee, West Virginia
ban during instructional time	2026-27	2	Michigan, Wisconsin
requires districts to adopt a policy	2024-25	1	Minnesota
requires districts to adopt a policy	2025-26	4	Alaska, Arizona, Idaho, New Mexico
requires districts to adopt a policy	2026-27	5	California, Colorado, Connecticut, Delaware, Wyoming
statewide bell-to-bell ban	2023-24	1	Florida
statewide bell-to-bell ban	2024-25	2	Louisiana, South Carolina

Policy type	Effective school year	N	States
statewide bell-to-bell ban	2025-26	14	Alabama, Arkansas, District of Columbia, Kentucky, Missouri, Nebraska, New Hampshire, New York, North Dakota, Ohio, Oklahoma, Oregon, Texas, Virginia
statewide bell-to-bell ban	2026-27	8	Georgia, Hawaii, Kansas, Maine, New Jersey, Rhode Island, Utah, Vermont
statewide bell-to-bell ban	2027-28	2	Illinois, Maryland

Compiled from Ballotpedia, Education Week, KFF, and enacted state statutes; classification differences across trackers are documented in the data file. Florida's HB 379 (instructional time, 2023–24) was the earliest statewide mandate.

Nearly all statewide phone bans took effect in fall 2024, fall 2025, or fall 2026. Because the latest mental-health survey waves currently in hand are from spring 2023 (YRBS, Pennsylvania, Maine), spring 2024 (Illinois), and fall 2025 (Washington, which has state guidance rather than a mandate), post-ban statewide mental-health outcomes are not yet observed in our panel. The upcoming 2025 YRBS state files will permit an initial difference-in-differences test comparing early-ban states—Florida (2023–24) and Indiana, Louisiana, and South Carolina (2024–25)—against states without a mandate across the 2021, 2023, and 2025 waves, by sex. As subsequent waves of the state county surveys are released, the county panel will test Minnesota's 2024–25 district-policy mandate, Maine's and California's 2026–27 mandates, and Illinois's 2027–28 bell-to-bell ban using the same county-by-grade panel structure developed for LTE above. Obtaining NCES restricted-use SSOCS files (which include state and county identifiers) would also enable a school-level panel design across the 2010–2024 relaxation and re-tightening cycle.

What the causal literature already shows

Study	Setting	Design	Outcome domain	Headline
Beland & Murphy (2016)	England	DiD across schools with staggered school-level bans (school survey on ban timing + NPD administrative data)	GCSE test scores	Post-ban test scores +6.41% of a SD; gains concentrated among lowest-achieving students (~+14% SD); no effect for high achievers (published: Labour Economics 41, 61-76)
Kessel, Hardardottir & Tyrefors (2020)	Sweden	DiD replication of Beland & Murphy with school survey on ban timing (~75% response) + register data	grade 9 test scores / GPA	No impact on student performance; 'can reject even small-sized gains' (Economics of Education Review 77, 102009)

Study	Setting	Design	Outcome domain	Headline
Beneito & Vicente-Chirivella (2022)	Spain	Synthetic control (Galicia) + DiD (Galicia, Castilla-La Mancha) vs other regions	PISA math/science; reported bullying cases	Galicia PISA math +10.7 and science +12.7 points by 2017 (0.6-0.8 and ~0.7-1 years of learning); CLM positive but insignificant; bullying cases per 10,000 down 15-18% (ages 12-14) and 9.5-18% (15-17), no effect under 12 (Applied Economic Analysis 30(90), 153-175)
Abrahamsson (2024)	Norway	Event study / staggered DiD using a survey of middle schools' smartphone policies linked to administrative health, education and family registers and the national pupil survey	GP and specialist consultations for psychological symptoms; reported bullying; GPA; academic high-school track	Girls: fewer GP visits for psychological symptoms (about -0.2 to -0.3 visits, a sizeable share of the mean) and ~60% fewer specialist mental-health visits; bullying down for both genders (girls exposed 3 years ~-46%); girls' GPA +0.08 SD and higher academic-track enrolment; effects larger for low-SES girls; no effects for boys (NHH DP 01/2024; forthcoming Journal of Human Resources per Allcott et al. 2026)
Goodyear, Randhawa, Adab, et al. (SMART Schools) (2025)	England	Cross-sectional comparison of schools with restrictive vs permissive recreational-phone policies (not causal)	WEMWBS mental wellbeing; anxiety; depression; sleep; attainment; phone/social-media time	No difference in wellbeing: adjusted mean WEMWBS difference -0.48 (95% CI -2.05 to 1.06, p=0.62); no differences in anxiety, depression, sleep, attainment or total daily phone/social-media use; less use during school hours only (Lancet Regional Health - Europe 51, 101211)
Figlio & Ozek (2025)	United States (Florida)	DiD comparing schools with high vs low pre-ban student phone activity (Advan building-level GPS) before/after the 2023 statewide ban, within one district	test scores (3x/year), disciplinary incidents, suspensions, absences	Year 1: incidents/suspensions +~25% raw, DiD +12% suspensions, in-school suspensions +~30% for Black male students, fading by year 2; year-2 test scores +0.6 percentiles overall, +1.1 percentiles on spring test (1.3 in middle/high, 1.4 for boys); unexcused absences significantly reduced in both years (NBER WP 34388)

Study	Setting	Design	Outcome domain	Headline
Allcott, Baron, Dee, Duckworth, Gentzkow & Jacob (2026)	United States (national)	Staggered DiD (Callaway-Sant'Anna) of Yondr lockable-pouch adoption using pouch sales records, teacher survey, GPS pings, state test scores and administrative data; pre-registered	phone use; test scores; attendance; discipline; student well-being; classroom attention; perceived online bullying	Phone use sharply down (teacher-reported personal use in class 61% -> 13%); test scores ~0 on average (rules out >0.007 student SD), high-school math +0.024 SD, middle school slightly negative; suspensions +~11% in year 1 then fade; subjective well-being -0.2 SD at adoption recovering to +0.16 SD by year 2; no effect on attendance, attention or online bullying (NBER WP 35132)
Lichand, Moreno-Louzada, da Costa & Gentzkow (2026)	Brazil (Rio de Janeiro)	DiD contrasting middle schools without strict pre-existing rules (treated) vs schools already strict (control) around the Aug 2023 municipal ban (school-wide from Feb 2024); principal survey + admin data	standardized test scores; in-school phone use	Test scores +0.06 SD in treated relative to control schools; in-school phone use fell substantially
Kohnstamm Instituut & Oberon (for Dutch Ministry OCW) (2025)	Netherlands	Descriptive monitoring: school-leader surveys and group interviews; no comparison group	leader-reported concentration, social climate, learning, well-being, enforcement	Secondary: 75% report better concentration, 59% better social climate, 28% better learning; share seeing non-educational phone use as a problem 62% -> 29%; some teachers report more physical bullying/disruption and enforcement workload. Primary: mostly no effect; 23% see better well-being/climate
OECD, PISA 2022 Results Volume II (ch. 5) and 'Managing screen time' (2024) (2023)	OECD/partner countries (cross-national)	Cross-sectional associations using PISA 2022 student/school questionnaires	self-reported distraction by digital devices; in-school phone use; notification habits; performance	About one in three OECD students attend schools banning phones; students in ban schools less likely to report distraction, but 29-30% of them still use a phone several times a day (+21% every day/almost) so bans are often unenforced; in 9 systems bans are concentrated in lower-performing schools; in some systems banned students are less likely to turn off notifications at night. No causal claim.

Study	Setting	Design	Outcome domain	Headline
Bottger & Zierer (2024)	Multi-country (review)	Rapid review + meta-analysis of 5 quantitative studies	social well-being (incl. bullying); academic performance	Pooled d = 0.162 (p<0.05); larger for social well-being than for achievement (Education Sciences 14(8), 906)
Campbell, Edwards, Pennell, et al. (2024)	Multi-country (review)	Scoping review	academic outcomes; mental health and wellbeing; cyberbullying	Mixed: 4 studies report achievement gains (2 restricted to low-achieving/low-SES samples); as many null or negative; no consistent effect on mental health or cyberbullying (J. Psychologists and Counsellors in Schools)

Across existing evaluations, evidence on school phone bans is mixed overall but points to girl-specific benefits when mental health is measured over multiple years. The strongest mental-health evidence comes from Abrahamsson's (2024) staggered difference-in-differences study of more than 400 Norwegian middle schools, which finds that bans reduced girls' general-practitioner and specialist visits for psychological symptoms, reduced bullying for both sexes, and raised girls' GPA, with no mental-health or achievement effect for boys. By contrast, an English cross-sectional study (Goodyear et al. 2025) finds no difference in student well-being between restrictive and permissive schools, and the 2026 U.S. Yondr-pouch study (Allcott et al.) finds an initial year-one dip in subjective well-being (−0.20 SD) followed by a year-two recovery (+0.16 SD) and no change in perceived online bullying. On academic achievement, Beland and Murphy (2016) in England, Beneito and Vicente-Chirivella (2022) in Spain, Figlio and Özek (2025) in Florida, and Lichand et al. (2026) in Rio de Janeiro find small positive test-score gains, whereas Kessel et al. (2020) in Sweden and Allcott et al. (2026) in the U.S. overall find near-zero average test-score effects (with modest high-school math gains in the U.S.).

County social-media use and sadness

State	County item	Outcome	County-grade-years	Across counties (SE)	Within county and grade (SE)
WA	social media daily	sad hopeless	172	+0.22 (0.04)	-0.09 (0.16)
WA	social media daily	considered suicide	172	-0.01 (0.03)	-0.15 (0.10)
WA	screen time 3plus	sad hopeless	172	+0.25 (0.06)	+0.01 (0.08)
WA	screen time 3plus	considered suicide	172	+0.03 (0.03)	+0.04 (0.07)
ME	screen time 3plus	sad hopeless	43	+0.07 (0.11)	-0.09 (0.10)

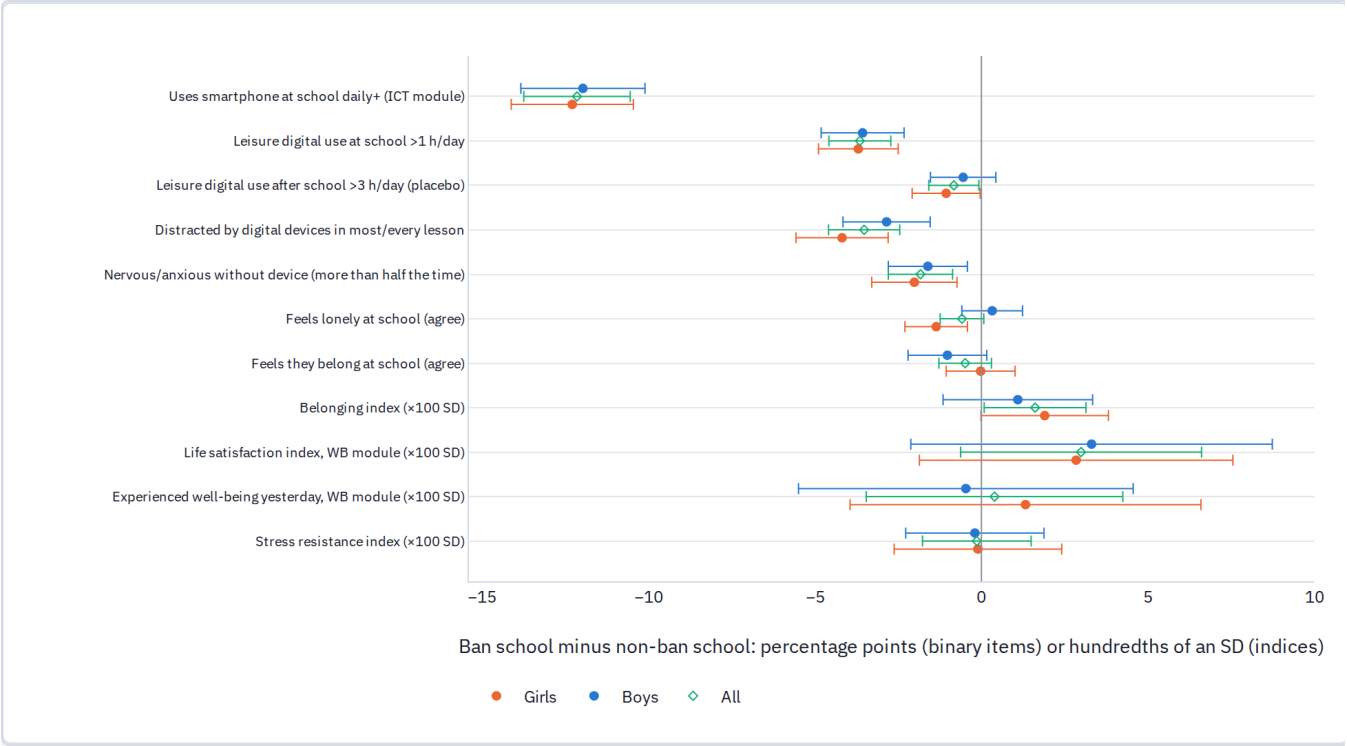
State	County item	Outcome	County-grade-years	Across counties (SE)	Within county and grade (SE)
ME	screen time 3plus	considered suicide	43	+0.22 (0.12)	+0.13 (0.13)

Percentage points of outcome per percentage point of the county screen-use item, estimated on county-by-grade cells so that exposure and outcome cover identical grades; "Across counties" includes year fixed effects, and "Within county and grade" includes county-by-grade and year fixed effects, with standard errors clustered by county. Washington covers grades 8, 10, and 12 in 2023 and 2025; Maine covers grades 9–12 in 2017, 2019, and 2021.

In Washington, county-grade cells where a higher share of students use social media daily also report higher rates of persistent sadness (+0.22 pp per percentage point, SE 0.04), and a similar cross-sectional relationship holds for three or more hours of daily screen time (+0.25 pp, SE 0.06). Between 2023 and 2025, both screen measures and sadness declined statewide in Washington (daily social media fell from 74.9% to 72.9%, 3+ hours of screen time from 68.9% to 65.5%, and sadness from 31.2% to 27.4%), but county-by-county changes over the two-year window do not line up: within the same county and grade, two-year changes in sadness are unrelated to changes in either screen-time measure (−0.09 pp, SE 0.16 for daily social media; +0.01 pp, SE 0.08 for 3+ hours of screen time). Maine's county panel from 2017 to 2021 likewise shows no within-county relationship. Just as in the national literature, the cross-sectional correlation between screen time and sadness is clear, while short-run within-area changes are noisy and null.

PISA 2022: school phone rules and well-being within countries

PISA 2022 asked school principals whether "the use of cell phones is not allowed on the school premises" and asked 15-year-old students about loneliness and belonging at school, digital-device habits, and (in 13 education systems) overall life satisfaction. Merging the student and school files yields 568,374 students across 19,885 schools in 80 systems; overall, 41% of students attend a school that bans phones on premises, ranging from 7% in Finland to 96% in Greece. We compare students in ban versus non-ban schools within the same country, controlling for student socio-economic status (ESCS), sex, grade, and private/public school status, using student survey weights and clustering standard errors by school.



Binary outcomes are in percentage points; PISA scale indices are in hundredths of a standard deviation. The ICT familiarity module (smartphone use at school) covers 51 systems; the well-being module (life satisfaction and experienced well-being) covers 13–14 systems.

Outcome	All	Girls	Boys	Mean
Uses smartphone at school daily+ (ICT module)	-12.15 (0.82)	-12.29 (0.94)	-11.97 (0.95)	55.3 pp
Leisure digital use at school >1 h/day	-3.65 (0.47)	-3.70 (0.61)	-3.57 (0.64)	33.4 pp
Leisure digital use after school >3 h/day (placebo)	-0.83 (0.38)	-1.06 (0.52)	-0.55 (0.50)	27.7 pp
Distracted by digital devices in most/every lesson	-3.52 (0.55)	-4.19 (0.71)	-2.85 (0.67)	28.1 pp
Nervous/anxious without device (more than half the time)	-1.83 (0.49)	-2.01 (0.65)	-1.61 (0.61)	22.5 pp
Feels lonely at school (agree)	-0.58 (0.33)	-1.36 (0.48)	+0.32 (0.46)	19.7 pp
Feels they belong at school (agree)	-0.49 (0.40)	-0.02 (0.53)	-1.02 (0.60)	77.5 pp
Belonging index (SD units)	+0.02 (0.01)	+0.02 (0.01)	+0.01 (0.01)	-0.2 SD
Life satisfaction index, WB module (SD units)	+0.03 (0.02)	+0.03 (0.02)	+0.03 (0.03)	-0.0 SD
Experienced well-being yesterday, WB module (SD units)	+0.00 (0.02)	+0.01 (0.03)	-0.00 (0.03)	0.1 SD
Stress resistance index (SD units)	-0.00 (0.01)	-0.00 (0.01)	-0.00 (0.01)	-0.1 SD

The within-country comparisons show three consistent patterns. First, school bans reduce in-school phone use, though compliance is far from complete: daily smartphone use at school is 12.1 percentage points lower in ban schools (from an overall mean of 55.3%, meaning roughly 43% of

students in ban schools still use a phone daily), leisure digital use of more than an hour per day at school falls by 3.7 points, and after-school leisure screen use (>3 hours/day) is nearly unchanged (−0.8 pp)—a useful check that ban schools are not simply attended by students who avoid screens outside school. Second, students in ban schools report less classroom distraction by digital devices (−3.5 pp overall; −4.2 pp for girls and −2.8 pp for boys). Third, well-being differences are small, favorable, and concentrated among girls: girls in ban schools are 2.0 pp less likely to feel nervous or anxious without their device, 1.4 pp less likely to report feeling lonely at school (versus +0.3 pp for boys), and score +0.02 to +0.03 SD higher on school belonging and life satisfaction. Because this is a cross-sectional comparison across schools, however, it cannot separate the causal effect of a phone ban from unobserved differences in school leadership or student composition.

Assessment. Across 80 education systems in PISA 2022, school phone prohibitions are associated with lower in-school phone use, less classroom digital distraction, and slightly lower loneliness and device anxiety among girls. While cross-sectional school comparisons cannot rule out selection into bans, the girl-specific pattern aligns with the quasi-experimental evidence from Norway's middle-school bans and underscores the value of evaluating the 2024–26 U.S. state phone bans as post-ban YRBS and state-survey waves become available.

How this sits with Haidt's and Twenge's evidence

The Anxious Generation research pages and collaborative reviews compile a large body of evidence linking smartphones and social media to adolescent distress, while Peter Gray and education researchers point to declining childhood autonomy, academic pressure, and the school calendar (Gray, Lancy, and Bjorklund 2023; Hansen, Sabia, and Schaller 2024). Much of the descriptive case for the phone hypothesis comes from Jean Twenge's work with the Monitoring the Future, YRBS, and Millennium Cohort surveys (correlating screen time with depressive symptoms and tracking national trends), paired with natural experiments on broadband, early social-media rollouts, and pre-2011 carrier exclusivity. Placing both literatures alongside our pilot results clarifies what public microdata can and cannot confirm: the individual-level association between heavy phone use and distress replicates readily in new settings, both camps converge on a long-run decline in teens' in-person socializing, and the sharpest quasi-experimental effects come from initial broadband or iPhone access and the school calendar rather than 2011–13 4G speed upgrades.

What that evidence base contains

Evidence type	Design	Typical result
Cross-sectional use–symptom associations (Twenge and colleagues)	Heavy vs. light users in national surveys (MTF, YRBS, Millennium Cohort, UK and U.S. panels)	Heavy users are roughly twice as likely to report depression or low well-being (adjusted odds ratios 1.7–3.0), with stronger associations among girls
Longitudinal panels	Screen or social-media use at wave t predicting symptoms at $t + 1$	Small effects; the main point of methodological debate with Orben, Przybylski, Odgers, and Ferguson
Experiments	Short-term deactivation and abstinence trials, mostly among college students and adults	Modest improvements of a few hundredths to roughly 0.2 SD
Broadband & social-media natural experiments	Facebook college rollout (2004–06); four European broadband IV studies—Germany (Golin 2022), Italy (Donati et al. 2025), Spain (Arenas-Arroyo et al. 2025), and the UK (McDool et al. 2020)—reviewed by McClean, Rausch, and Haidt; U.S. state broadband against YRBS (Churchill and Johnson 2026); AT&T's 2007–11 iPhone carrier monopoly (Myers and Hooper 2026); and Elaine Guo's British Columbia wireless study (2023)	Facebook arrival: ~0.1 SD worse on a mental-health index. Italy DSL (2001–13): +0.08 SD in hospital-diagnosed disorders among youth. Spain fiber-to-the-home (2007–19): significant rise in adolescent hospital mental-health and self-harm diagnoses, driven by girls. U.S. broadband × YRBS (2009–19): +9–17% suicidal ideation per SD of broadband access, concentrated among girls. AT&T 2007–11 iPhone monopoly: crowded out teen in-person socializing and reduced teen birth rates by 4.5–8.0%. British Columbia fixed wireless (2014+): +1 pp (~90% of baseline) in girls' severe school mental-health designations
School calendars, closures & declining autonomy (Gray and colleagues)	Monthly NVSS youth suicide seasonality across the school calendar and county school closures/reopenings in 2020–21 (Hansen and Lang 2011; Hansen, Sabia, and Schaller 2024); synthesis of long-run declines in children's unsupervised play and independent mobility (Gray, Lancy, and Bjorklund 2023)	Adolescent suicides (ages 12–18, unlike 19–25) drop sharply every summer and winter break, fell 18–25% during spring 2020 pandemic school closures despite record screen time, and rose again as counties reopened in-person schooling. Multi-decade declines in independent play and teen face-to-face socializing (also emphasized in Part II of *The Anxious Generation*) predate smartphones and continued through the 2010s
Aggregate timing	The mood-disorders review and the World Happiness Report chapter	Flat trends through the Great Recession, an inflection around 2012 led by girls across the Anglosphere and Western Europe, and declining PISA life satisfaction in 40 of 47 countries between 2015 and 2018

Where the pilot agrees

- In the 2022–23 GSHS rounds across five countries, girls who use a mobile phone three or more hours a day report loneliness 8 percentage points higher and suicidal ideation 10 points higher than girls who use a phone less, with rates rising steadily at higher hours. That corresponds to an odds ratio of roughly 1.5 to 1.7, similar to the cross-sectional estimates in U.S. and European surveys.
- In the 2022 Nepal DHS, self-reported PHQ-9 and GAD-7 scores among 15–19-year-old women show a positive, though imprecisely estimated, within-cluster association with daily internet use.
- The national U.S. trends line up across sources: NSDUH youth depression turns upward in 2011–12 and YRBS sadness follows, the increases are consistently larger for girls than for boys, and the ACS cognitive-difficulty rate among adults 65 and older falls over the same decade. In ATUS time diaries, weekday homework time rose by roughly 40% while teens' in-person socializing fell—consistent with both Gray's emphasis on rising academic structure and Haidt's account of declining face-to-face peer time.
- In state-level panels, the NTIA share of teen households using smartphones for internet access predicts YRBS girls' sadness (+0.9 pp per 10 percentage points, SE 0.44). That aligns directionally with Churchill and Johnson (2026), who link FCC state broadband expansion to the 2009–2019 YRBS and find a 9–17% increase in adolescent suicidal ideation per standard deviation of broadband access (concentrated among girls alongside cyberbullying and body dissatisfaction), though in our county-level youth survey panels the adoption relationship does not hold within states.

Where the pilot finds no effect, and how to interpret the difference

1. **LTE rollout at the metro and county levels.** Most of the natural experiments cited in the *Anxious Generation* literature estimate the effect of *first* high-speed broadband access (Donati et al. 2025; Arenas-Arroyo et al. 2025), of Facebook first arriving on college campuses, or of early U.S. smartphone exclusivity. On that last point, Myers and Hooper (2026) show that during AT&T's exclusive U.S. carrier monopoly on the iPhone from June 2007 to February 2011, pre-existing county-level AT&T 3G coverage strongly predicted early iPhone adoption, crowding out teens' in-person socializing and reducing 15–19 birth rates by 4.5–8.0%. By the time commercial 4G LTE rolled out across U.S. metros in 2011–2013—and after Verizon began carrying the iPhone in February 2011—3G smartphones and mobile social media were already spreading nationwide, and most large metros received LTE within a narrow two-year window. Finding a null effect for 4G LTE launch dates therefore shows that 2011–13 metro speed upgrades are too late and compressed to isolate the U.S. smartphone transition, whereas pre-2011 AT&T 3G coverage and county broadband availability offer sharper variation.

   The closest post-2010 geographic parallel that does find a mental-health effect is Elaine

Guo's British Columbia study (alongside Arenas-Arroyo et al.'s Spanish fiber-to-the-home study using hospital records). Guo compares teenagers in rural mountain communities that gained high-speed fixed wireless internet (25 Mbps over cellular infrastructure) with teenagers in mountain communities that remained unserved, comparing girls with boys before and after 2014 using administrative school records on severe mental-health designations (which require a clinical diagnosis). Starting in 2014, girls in covered communities experienced a roughly 1 percentage point increase in severe designations relative to boys—about 90% of their baseline rate—with no corresponding shift for boys, moderate designations, or unrelated disabilities. Three features distinguish Guo's setting from our U.S. LTE panel: the treatment is a transition from little or no home broadband to high-speed access in isolated towns rather than a 3G-to-4G upgrade in metropolitan areas; the outcome is a clinical school designation rather than a census disability screener; and the time contrast compares pre- and post-2014 periods against a 2019 coverage map rather than staggered rollout dates (which is also a limitation of that study, since earlier annual coverage maps were unavailable). Taken together, these studies suggest that geographic designs detect a 2010s connectivity effect when they capture initial high-speed access or pre-2011 carrier exclusivity and measure clinical or self-reported internalizing outcomes.

2. **Country-level GSHS panels with national internet, mobile-broadband, and 3G/4G series.** Regressing country-round survey means on national ITU penetration series is subject to the same aggregation problems that Haidt and coauthors highlight in Vuorre and Przybylski's 72-country Facebook study: national averages pool adult and youth access and move with broader economic trends. Our country-level GSHS regressions should be read with the same caution.
3. **Caregiver reports in 67 MICS surveys.** The MICS analysis provides a more substantive null result. Across 507,478 children in low- and middle-income countries, caregiver-reported anxiety and depression among 10–17-year-olds show no meaningful relationship with local young-adult internet use or with a teenage girl's own daily phone or internet use within the same sampling cluster; the small positive cluster correlation for girls' anxiety is mirrored among 5–9-year-olds who do not use phones. Two interpretations remain open: either the self-report correlations in school surveys partly reflect reverse causation and shared reporting style (distressed adolescents both use phones more and report more symptoms), or caregivers frequently miss internalizing symptoms in their children (consistent with clinical evidence that parent–child agreement on internalizing disorders is low). Public cross-sections in MICS alone cannot separate these two explanations, but they can be tested directly in the U.S. by linking parent and teen reports in NHIS-Teen and internationally by expanding the DHS-8 self-report mental-health panel.

Two empirical patterns the existing debate has largely overlooked

- Perceived schoolwork pressure rose after 2010 across roughly 45 HBSC countries, not only in the United States, and psychosomatic complaints rose more in countries where pressure increased more. Although the within-country slope is modest and matches the pre-2010

baseline relationship, the post-2010 rise in perceived academic pressure is a broad cross-national trend that dismissals based on single-country exam reforms do not address. Combined with Hansen, Sabia, and Schaller's (2024) finding that U.S. youth suicides drop sharply every summer, fell by 18–25% during the spring 2020 pandemic school closures despite surging screen time, and rebounded when in-person schooling resumed, the evidence points toward an interaction (H3): digital peer comparison and notifications are most harmful when layered onto the social and academic stressors of the school year.
- In U.S. state youth surveys with post-2021 waves, adolescent sadness and suicidal ideation have fallen substantially from their 2021 peaks—in Washington, 2025 sadness is below its 2010 rate—without a corresponding drop in daily smartphone or social-media use. Neither a simple screen-dose account nor a simple academic-pressure account explains that downturn without additional mechanisms, such as the re-tightening of school phone restrictions or shifts in platform content.

Summary. Prior literature documents a consistent individual-level correlation between heavy screen use and adolescent distress alongside causal effects from initial broadband rollouts (Donati et al. 2025; Arenas-Arroyo et al. 2025; Churchill and Johnson 2026; Guo 2023), early iPhone carrier exclusivity (Myers and Hooper 2026), and the school calendar (Hansen, Sabia, and Schaller 2024). Our pilot reproduces the individual-level association in developing-country school surveys, but finds that 2011–13 U.S. 4G launch dates, county smartphone-adoption rates, and caregiver reports in 67 MICS surveys show no corresponding effect. Resolving where the causal effect lies requires pairing self-reported and clinical outcomes with sharper local variation—including the 2024–26 U.S. school phone bans, pre-2011 carrier coverage, school-year seasonality, and linked parent–teen surveys.

Conclusions and next steps

Combining public microdata across the United States and more than 80 other countries narrows the debate in several ways while leaving the core causal question open.

Hypothesis	Where it stands after the pilot
H1 · Phones and social media	The cross-sectional individual association is robust and extends to developing countries: teenage girls who use mobile phones heavily report substantially higher loneliness and suicidal ideation in Ghana, India, Jamaica, Morocco, and Nepal just as in Western surveys, with the same sex asymmetry and dose-response gradient. Yet every test in our pilot based on 2011–13 rollout timing or adult- and cluster-measured exposure is null: 4G launch dates across U.S. metros and counties, metro adoption rates in the county panels, cluster connectivity and own use in 67 MICS caregiver surveys, and national 3G/4G coverage in 34 GSHS countries. Published natural experiments that do find causal effects exploit earlier transitions—initial fixed or wireless broadband (Donati et al. 2025; Arenas-Arroyo et al. 2025; Churchill and Johnson 2026; Guo 2023) or AT&T's 2007–11 U.S. iPhone carrier monopoly (Myers and Hooper 2026)—rather than 3G-to-4G speed upgrades. School phone bans in PISA 2022 and Norway provide modest evidence of a girl-specific benefit.
H2 · School pressure	The descriptive premise of Gray's account holds up better than critics often acknowledge: perceived schoolwork pressure rose after 2010 across most HBSC countries, U.S. weekday homework time rose by roughly 40% while teens' in-person socializing fell (Gray, Lancy, and Bjorklund 2023), and independent evidence shows that U.S. youth suicides drop every summer and fell 18–25% during the spring 2020 pandemic school closures (Hansen, Sabia, and Schaller 2024). Within our panels, however, perceived pressure behaves like a steady correlate of psychosomatic complaints (with the same within-country slope before and after 2010) rather than the sole driver of the post-2010 break; U.S. state accountability reforms show no effect; and the ACS age gradient does not peak in the high-stakes high-school years.
H3 · Interaction	State accountability policies interacted with metro 4G launch dates show no positive effect on the ACS item, but the school-day version of the interaction hypothesis—smartphones operating inside high-pressure school hours—is consistent with both the school-calendar literature and the school-ban evidence: youth suicides track the in-person school calendar (Hansen, Sabia, and Schaller 2024), school bans reduce daily in-school phone use and classroom distraction, girls in ban schools report lower loneliness and device anxiety, the loneliness reduction is directionally larger among students with high mathematics anxiety, Norway's middle-school bans reduced girls' psychological consultations, and U.S. adolescent sadness moved inversely to national school phone restrictions.
H0 · Measurement	Ruled out as the primary explanation: the post-2010 rise appears simultaneously across diagnostic interviews, self-report school surveys, and the ACS cognitive-difficulty item; the 65+ comparison group trends downward over the same decade; and the increase is consistently concentrated among girls. Differential awareness and diagnosis-seeking likely contribute to the steeper relative rise among affluent and private-school teens, but do not account for the overall trend.

Three empirical findings from the pilot stand out beyond the existing book-length arguments. First, the 2010s rise in adolescent distress occurred almost simultaneously across U.S. states and counties with very different 4G-rollout and school-policy timelines, which explains why coarse geographic designs struggle to separate the competing hypotheses and why national time series can be cited in support of either view. Second, post-2021 waves of the U.S. state youth surveys show a substantial decline in sadness and suicidal ideation—returning to or below 2010 levels in Washington and Illinois—without a matching drop in daily phone or social-media use, coinciding instead with the re-tightening of school phone rules. Any purely dose-based account of screen time or school pressure has to account for that downturn. Third, across developing-country surveys, the link between phone use and distress appears consistently when adolescents report both their own hours and their own symptoms (GSHS, Nepal DHS), but vanishes when symptoms are

reported by caregivers (67 MICS surveys). Resolving why self-reports and caregiver reports diverge is central to the next phase of work.

Priority next steps for Phase 2 (detailed in `notes/phase2_plan.md`).

- **Resolve the caregiver vs. adolescent self-report gap in public data:** Link the public-use **NHIS-Teen** and **NHIS Sample Child** files (2021–2024)—which administer the MICS/Washington Group parent items and teen self-reported PHQ-2/GAD-2 and screen-time items to the *same* 12–17-year-olds—alongside the **National Survey of Children's Health (NSCH)** and additional **DHS-8 mental-health modules** (Mozambique 2022–23, Jordan 2023, Lesotho 2023–24, Cambodia 2021–22).
- **Test school-calendar seasonality and student-level homework × screen interactions:** Use public **CDC WONDER / NVSS monthly mortality series (1999–2024)** to test whether the post-2010 rise in adolescent suicide occurred primarily during the school year (September–May, as H2 and H3 predict following Hansen, Sabia, and Schaller 2024) or equally during the summer months (June–August). Pair this with **Monitoring the Future (MTF)** public microdata (1991–2024) and the **multi-wave PISA panel (2012, 2015, 2018, 2022, including PISA 2015's schoolwork-anxiety scale `ANXTEST`)** to estimate student-level interactions between homework/academic anxiety, declining in-person socializing, and screen time.
- **Evaluate the 2023–27 U.S. school phone bans and earlier broadband/carrier instruments:** Add the posted 2025 Pennsylvania (PAYS), 2019–23 California (CHKS), and 2022/2025 Minnesota (MSS) county tables, and track the 2023–27 state and district phone bans through the 2025–27 YRBS and county youth survey waves across at least two post-ban years. In the historical panels, replace metro 4G LTE launch dates with county-level **AT&T 2007–11 iPhone carrier-monopoly coverage** (Myers and Hooper 2026) and **NTIA/FCC county broadband availability maps (2010–2018)**.
- **Targeted restricted/licensed data (where public variation is insufficient):** License GSMA mobile-coverage polygons with MICS and DHS cluster coordinates for the international self-report modules, and request student-level HBSC microdata to test within-school interactions between problematic social-media use, schoolwork pressure, and school phone rules. Further extensions of the ACS market event study or the IPUMS International census cognitive-difficulty item are not warranted.

Reproducibility: analysis scripts are in `scripts/` and output tables are in `results/tables/` of the project repository. Data sources: IPUMS USA (ACS 1-year 2005–2024), IPUMS ATUS 2003–2024, IPUMS MICS (68 surveys), IPUMS International (Kenya 2009/2019, Mexico 2010/2020), Census CPS Computer and Internet Use Supplements (2009–2023), OECD PISA 2022, WHO GSHS (165 country-rounds), WHO HBSC, DHS-8

(Nepal 2022, Kenya 2022), and public NSDUH, NHANES, YRBS, NCES SSOCS/School Pulse, and state youth survey files as documented in each section.